Analysis of Emissions

STT461

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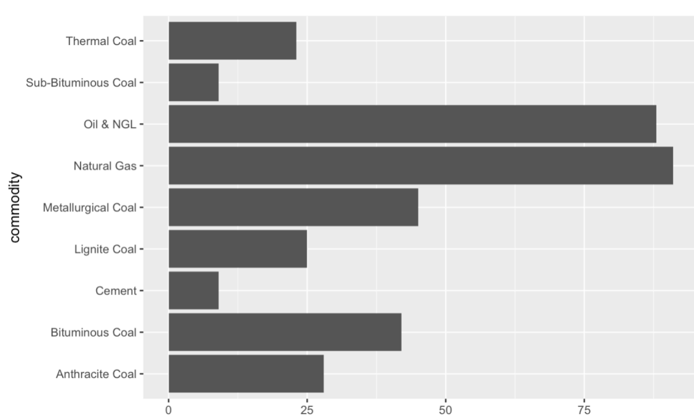
1. Introduction

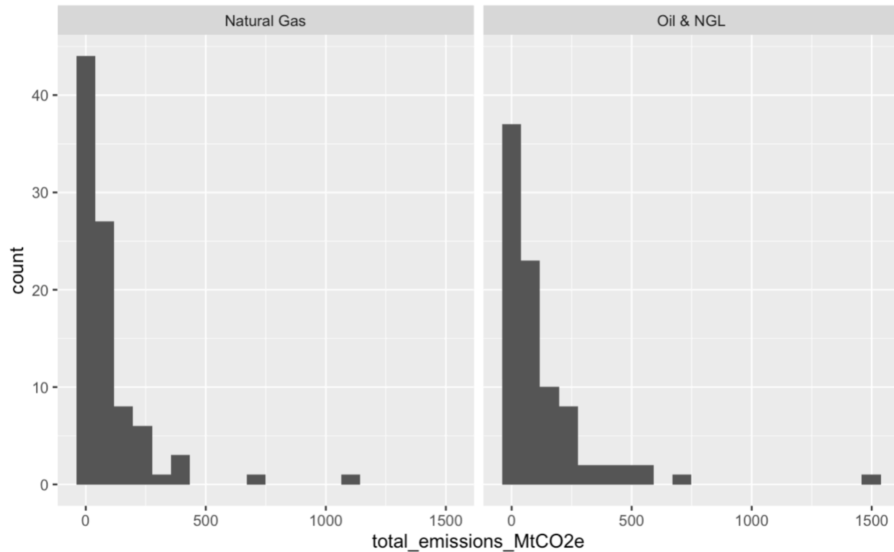
In our project we analyzed the use of fossil fuels, production of fossil fuels, and the leakage of fossil fuels in different plants based on Natural Gas or Oil & NGL and State vs Privately owned entities. Fossil fuel use and climate change is an ever growing and changing issue in our society today. With our Earth getting warmer and habitats changing it is important to analyze where some of these emissions are coming from. In our project we explore whether the total emissions from Natural Gas plants differ from Oil & NGL plants. We also construct confidence intervals of total emissions for both types of plants. We also test whether there is a significant difference in fugitive methane emissions(how much leaks out of the facility) as well as a 95% confidence interval for the difference between the 2. We investigate if the mean total emissions are significantly different for state owned entities vs privately owned entities. Lastly we investigate a Naive Bayes estimator to try and classify the plant as Natural Gas or Oil & NGL based on total mean CO2 emissions and fugitive methane emissions.

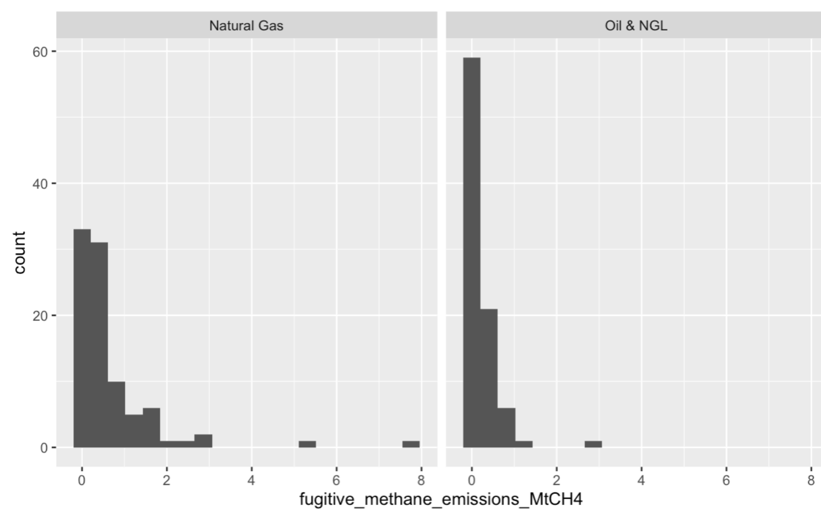
2. Data information

The data comes from the Carbon Majors Dataset from carbonmajors.org. This project traces fossil fuel emissions back to the producing plants of them. This dataset is updated yearly with the new information for each plant. There are 3 different granulites each with varying amounts of info. For our project we used the high granularity dataset which had the most information. There were many variables in the dataset including year, parent type, commodity, product emissions, own use emissions, fugitive methane emissions, and total emissions. The original dataset was 20,054 x 17 or 20,054 data points and 17 variables. Since the dataset had multiple years of the same companies all of the entries were not considered independent and therefore lots of our methodology could not be run. We filtered to just data points in the year 2024 to model an independent sample. For our project we also only focused on the variables commodity, parent type, total emissions, fugitive emissions, own fuel use emissions, and year. This filtered dataset was 360 x 6. There is a reporting identity column which leads us to believe the data is either self reported by the company or a third party company. A couple plots that fueled our investigations were



From this plot we saw that Natural Gas and Oil & NGL plants had the most data points so we decided to focus more on those 2.





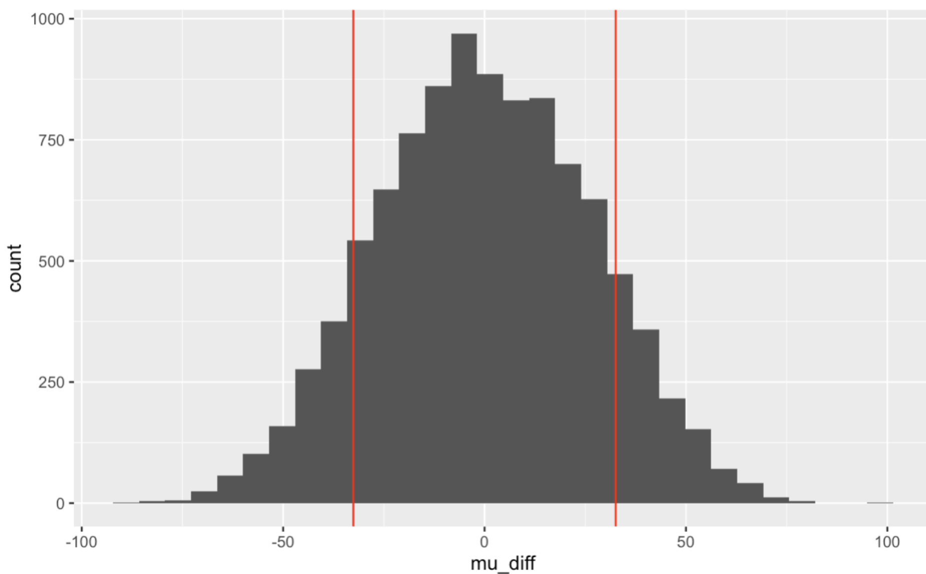
These plots or total emissions and fugitive methane emissions gave us some idea of what future tests to run and what the results may look like

3. Analysis

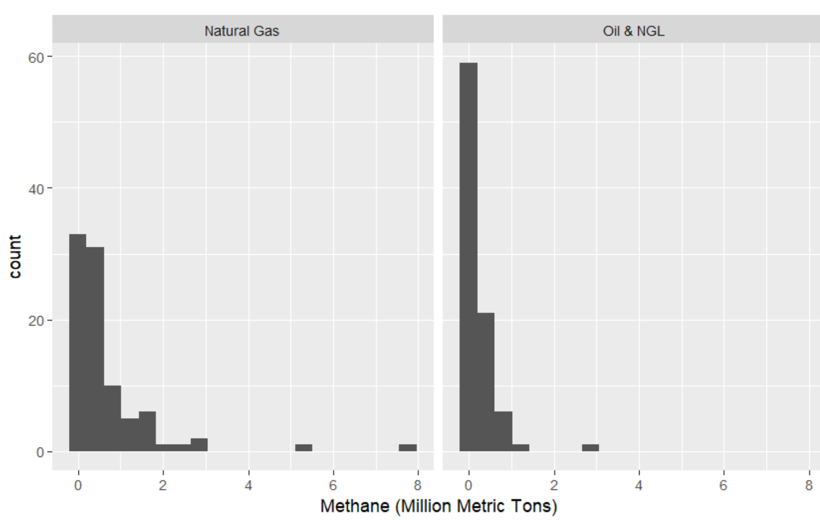
3.1 Are the mean total emissions of Natural Gas plants significantly different from Oil & NGL plants?

H0: $\mu NG = \mu Oil\&NGL$, HA: $\mu NG \neq \mu Oil\&NGL$, significance level, $\alpha = 0.05$

The plot that motivated this question is shown in Image 2 above.
In order to determine whether the mean total CO2 emissions of Natural Gas plants is significantly different from Oil & NGL plants, we replicated 10000 samples to compare the mean of the two groups. The p-value was calculated by taking the mean amount of times that the simulated differences were greater than the observed differences in sample means, then multiplying by 2 to account for each end. We obtained a p-value of .2286, which shows that there is little evidence against the null hypothesis that the mean amount of total CO2 emissions is the same for both natural gas plants and oil & ngl plants. We can not conclude that the total CO2 emissions significantly differ between the two plants.

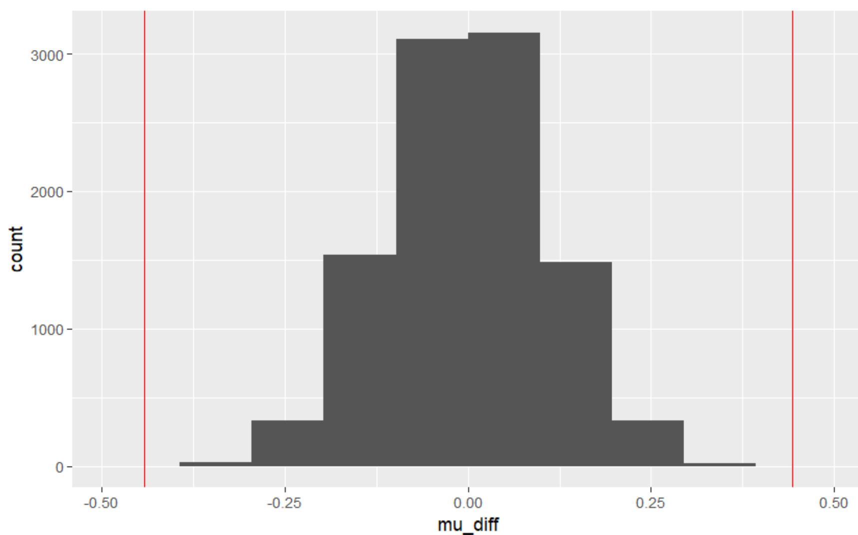


3.2 Are the mean fugitive methane emissions of Natural Gas plants significantly different from Oil & NGL plants?

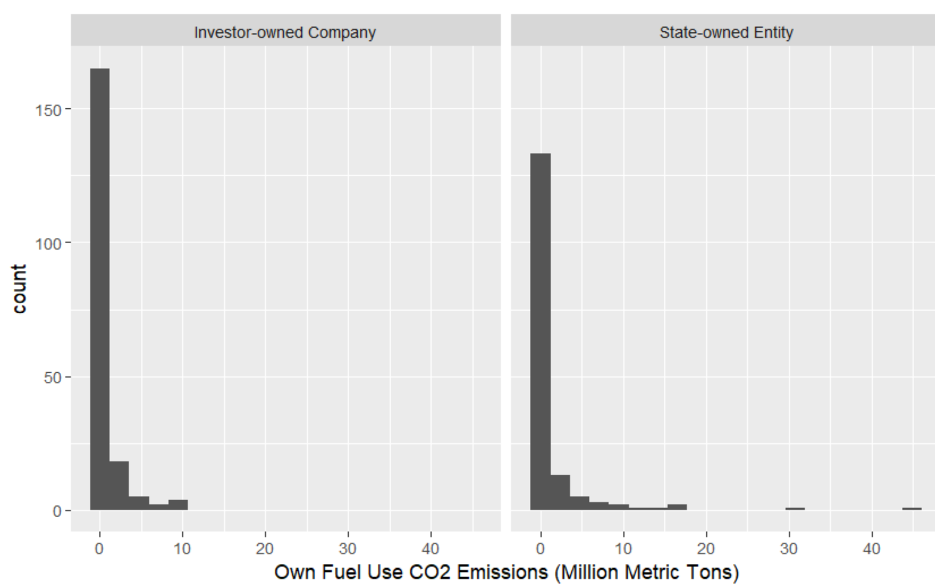


H0: μNG = μOil&NGL, HA: μNG ≠ μOil&NGL, significance level, α = 0.05

In order to determine whether the mean fugitive methane emissions of Natural Gas plants is significantly different from Oil & NGL plants, we replicated 10000 samples to compare the mean of the two groups. The p-value was calculated by taking the mean amount of times that the simulated differences were greater than the observed differences in sample means, then multiplying by 2 to account for each end.
We obtained a p-value of 0, which shows that there is very strong evidence against the null hypothesis that the mean amount of fugitive methane emissions is the same for both natural gas plants and oil & ngl plants. Fugitive methane emissions significantly differ between the two plants.

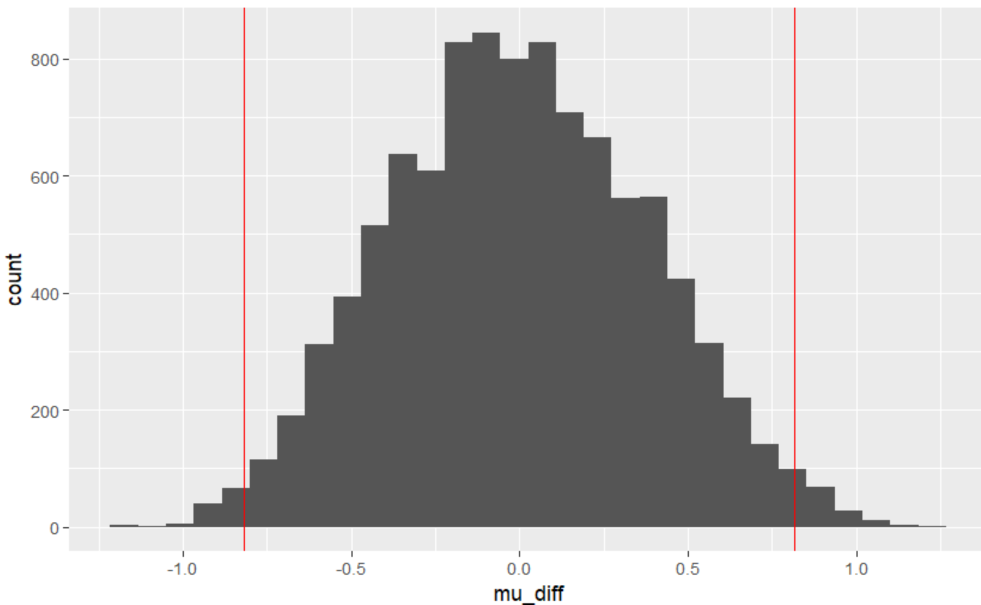


3.3 Using a permutation test, determine whether the mean own fuel use emissions differ significantly between state-owned and investor-owned

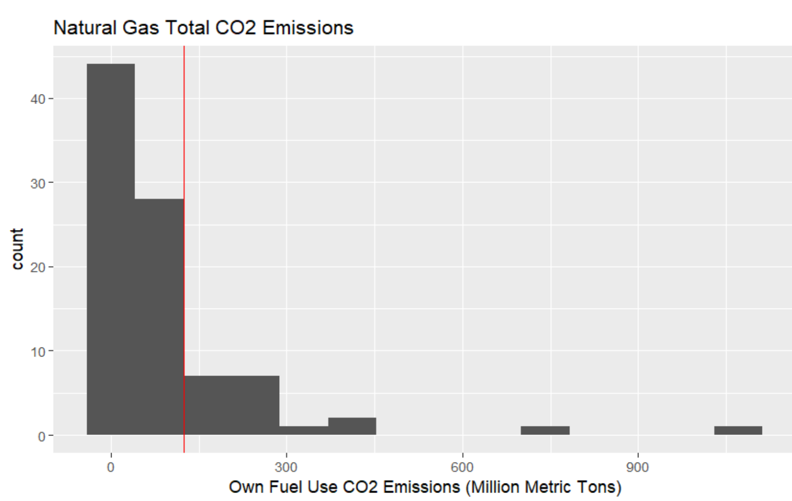


H0: μState = μInvestor,HA: μState ≠ μInvestor, significance level, α = 0.05

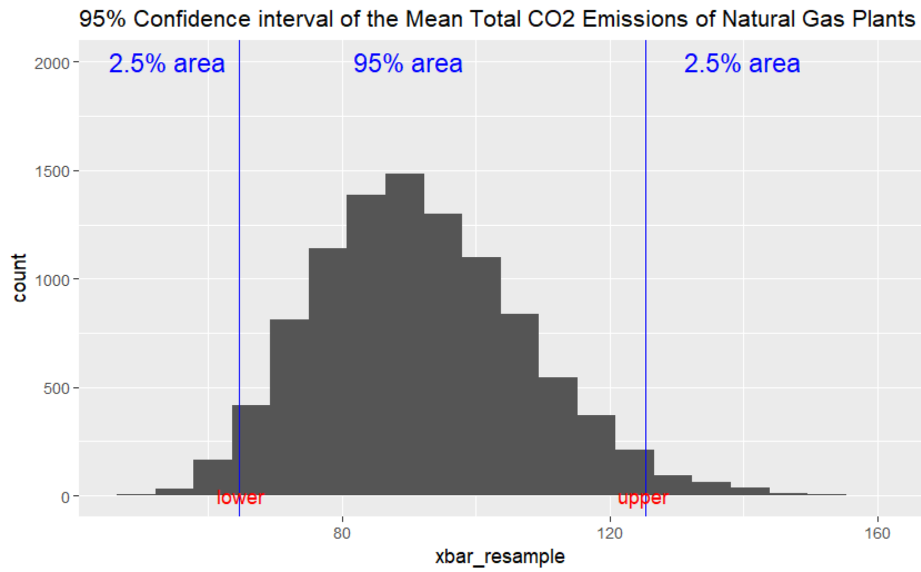
In order to determine whether the mean own fuel use CO2 emissions of state-owned entities is significantly different from investor-owned companies, we replicated 10000 samples to compare the mean of the two groups. The p-value was calculated by taking the mean amount of times that the simulated differences were greater than the observed differences in sample means, then multiplying by 2 to account for each end.
We obtained a p-value of 0.024, which shows that there is somewhat strong evidence against the null hypothesis that the state-owned mean own fuel use emissions are the same as investor-owned mean own fuel use emissions. State-owned mean own fuel use emissions are statistically different from investor-owned mean own fuel use emissions.



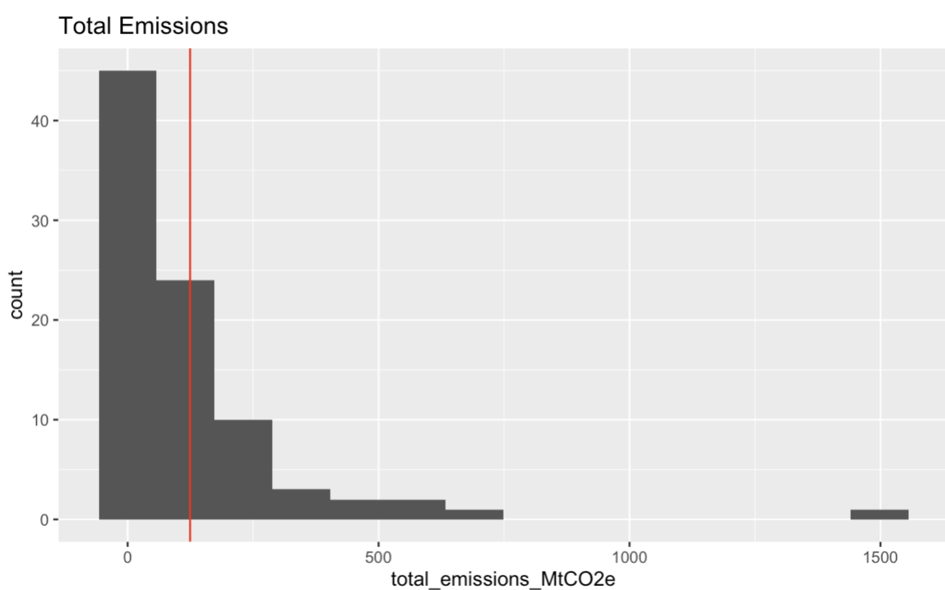
3.4 What is the 95% confidence interval for the mean total emissions of Natural Gas plants?



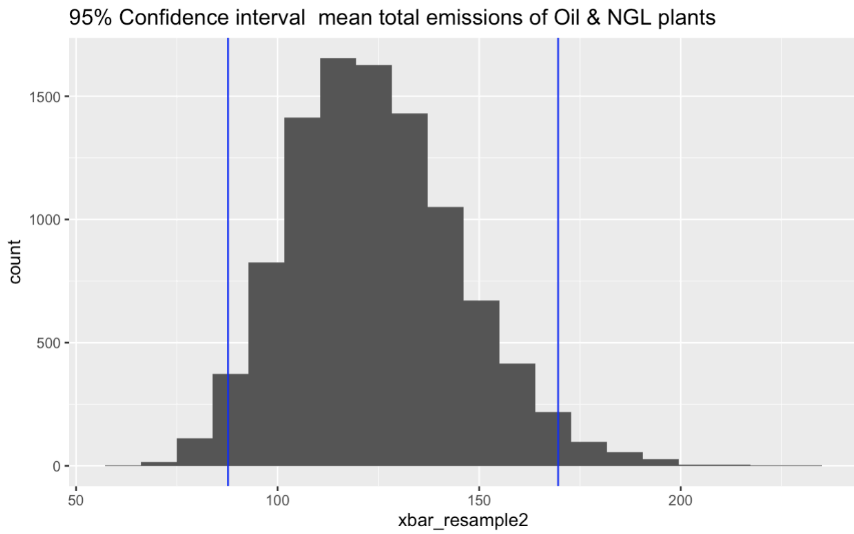
We used bootstrap resampling using 10000 replicates to estimate the sampling distribution of CO2 emissions. To find the 95% confidence interval, we used the quantile function on the resampled mean to find the 2.5th percentile and 97.5th percentile values. The 95% confidence interval was found to be [64.656, 125.362]. The confidence interval calculated allows us to estimate the mean total CO2 emissions of Natural Gas plants. Specifically, we are 95% confident that the mean total CO2 emissions of Natural Gas plants is between 64.656 and 125.362 million metric tons.



3.5 What is the 95% confidence interval for the mean total emissions of Oil & NGL plants?

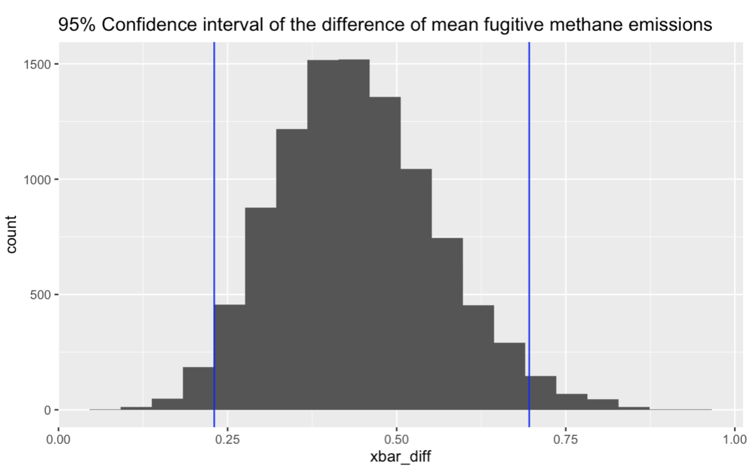


We used bootstrap resampling using 10000 replicates to estimate the sampling distribution of CO2 emissions. To find the 95% confidence interval, we used the quantile function on the resampled mean to find the 2.5th percentile and 97.5th percentile values. The 95% confidence interval was found to be [87.69, 169.5783]. The confidence interval calculated allows us to estimate the mean total CO2 emissions of Oil & NGL plants. Specifically, we are 95% confident that the mean total CO2 emissions of Oil & NGL plants is between 87.69 and 169.58 million metric tons.



3.6 What is the 95% confidence interval for the difference of mean fugitive methane between Natural Gas Plants and Oil & NGL?

We used bootstrap resampling using 10000 replicates to estimate the sampling distribution of the distribution of the difference in fugitive methane emissions between Natural Gas and Oil & NGL plants.. To find the 95% confidence interval, we used the quantile function on the resampled mean to find the 2.5th percentile and 97.5th percentile values. The 95% confidence interval was found to be [.2302,.6960]. The confidence interval calculated allows us to estimate the mean fugitive emissions of Oil & NGL plants.
Specifically, we are 95% confident that the mean fugitive emissions of Oil & NGL plants is between .2302 and .6960 million metric tons. The absence of 0 within the interval gives us more evidence that natural gas plants have more leakage than Oil & NGL plants.



3.7 Given an entity's total CO2 emissions or fugitive methane emissions, can you predict whether it came from a Natural Gas or Oil & NGL plant?

We trained two Naive Bayes classifiers using 5-fold cross validation. One of the classifiers is for CO2 emissions, and the other is for fugitive methane emissions.
The CO2 model resulted in a kappa of value of 0.032, which shows that there was poor agreement between the true and predicted values, so it was not a reliable classifier. The fugitive methane model resulted in a kappa value of 0.207, which shows that there is fair agreement between the true and predicted values, so this classifier has some slight predictive power. Methane is a more reliable predictor than CO2 when classifying between natural gas and oil & ngl plants. Finally we made a multiple variable classifier with CO2 emissions,fugitive methane, and parent type. This classifier refused to take the parent type for predictions so it was really just the CO2 emissions and methane emissions. On the testing data we saw an accuracy of .7143 and a kappa of .4186 showing moderate agreement and decent accuracy. Looking more into the confusion matrix we see that when the plant was Oil & NGL it was able to predict it every time, but when the plant was Natural Gas it only predicted Natural Gas 7/17 times.

```
Confusion Matrix and Statistics

             Reference
Prediction    Natural Gas Oil & NGL
  Natural Gas           7         0
  Oil & NGL            10        18

               Accuracy : 0.7143
                 95% CI : (0.537, 0.8536)
    No Information Rate : 0.5143
    P-Value [Acc > NIR] : 0.013006

                  Kappa : 0.4186
```

4.Conclusions

Our hypothesis testing revealed that the mean total CO2 emissions are statistically similar between Natural Gas and Oil & NGL plants, having a p-value of 0.2286, though the fugitive methane emissions are very different, with a p-value of 0. This shows that the environmental impact by these facilities should not only be measured by their CO2 emissions, the type of greenhouse gas being emitted varies by the commodity. When we analyzed whether the mean own fuel use CO2 emissions were different between state-owned entities and investor-owned companies, we found that they significantly differ, obtaining a p-value of 0.0244. This means the type of owner can have an influence on the amount of mean own fuel use CO2 being emitted.
We were successfully able to find 95% confidence intervals for the mean total CO2 emissions of Natural Gas plants, mean total CO2 emissions of Oil & NGL plants, and for the difference of mean fugitive methane emissions. The corresponding confidence intervals, found by using bootstrap resampling, are [64.656, 125.362] million metric tons, [87.696, 169.578] million metric tons, and [0.23, 0.696] million metric tons.

While trying to classify a plants type on their total emissions of CO_2 was shown to be unreliable with a kappa value of 0.032, using fugitive methane emissions as a predictor actually showed to have some predictive power, showing fair agreement between the true and predicted values with a kappa value of 0.207. We found that fugitive methane emissions were a more accurate, better predictor of a plant's type. Combining the two into one classifier gave us an accuracy of .7146 and a kappa of .4186 showing that it was a decent model at predicting the plant commodity type.

This research shows the complexity regarding the emissions of greenhouse gases by these entities and companies. Since emissions can vary between the plant type and ownership, there should be environmental policies to track specific gases and types of company in order to combat climate change.

Project461

Tyler Boerkoel and Seth Norder

2026-04-10

```r
library(tidyr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)
library(caret)
```

```
## Loading required package: lattice
```

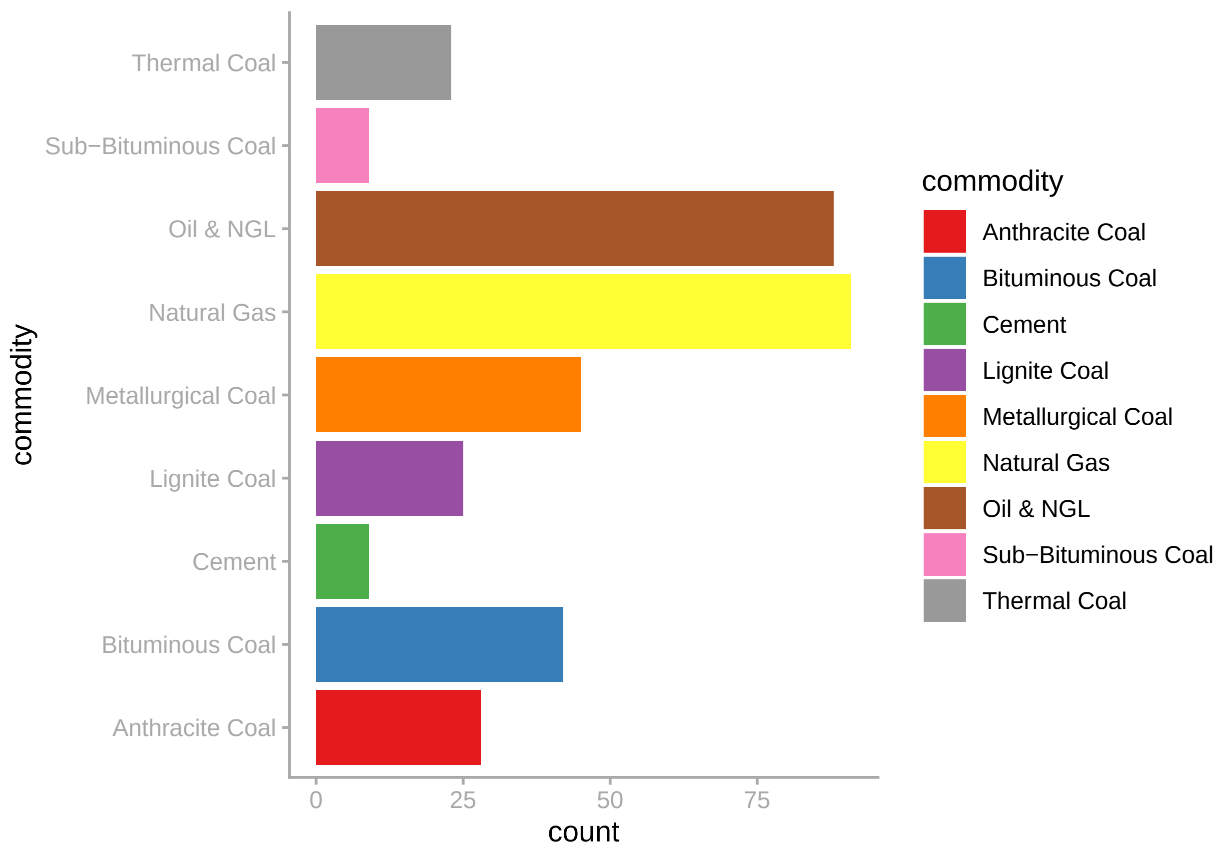
```r
emissions_data <- read.csv("emissions_high_granularity.csv")
#emissions_data <- read.csv("C:/Users/tyler/Downloads/emissions_high_granularity.csv")

emissions_data <- emissions_data%>%drop_na(commodity,
                                           parent_type,
                                           total_emissions_MtCO2e,
                                           fugitive_methane_emissions_MtCH4,
                                           own_fuel_use_emissions_MtCO2,
                                           )
emissions_data2 <- emissions_data[c("commodity",
                                    "parent_type",
                                    "total_emissions_MtCO2e",
                                    "fugitive_methane_emissions_MtCH4",
                                    "own_fuel_use_emissions_MtCO2",
                                    "year")]
emissions_data_24= emissions_data2 %>%
  filter(year == "2024")

data_ng_oil=emissions_data_24 %>%
  filter(commodity %in% c("Natural Gas", "Oil & NGL"))
```

Exploratory Data Anaylsis

```
table(emissions_data_24$commodity)
```

```
##
##     Anthracite Coal     Bituminous Coal              Cement       Lignite Coal
##                  28                  42                   9                 25
##   Metallurgical Coal         Natural Gas          Oil & NGL Sub-Bituminous Coal
##                  45                  91                  88                  9
##        Thermal Coal
##                  23
```

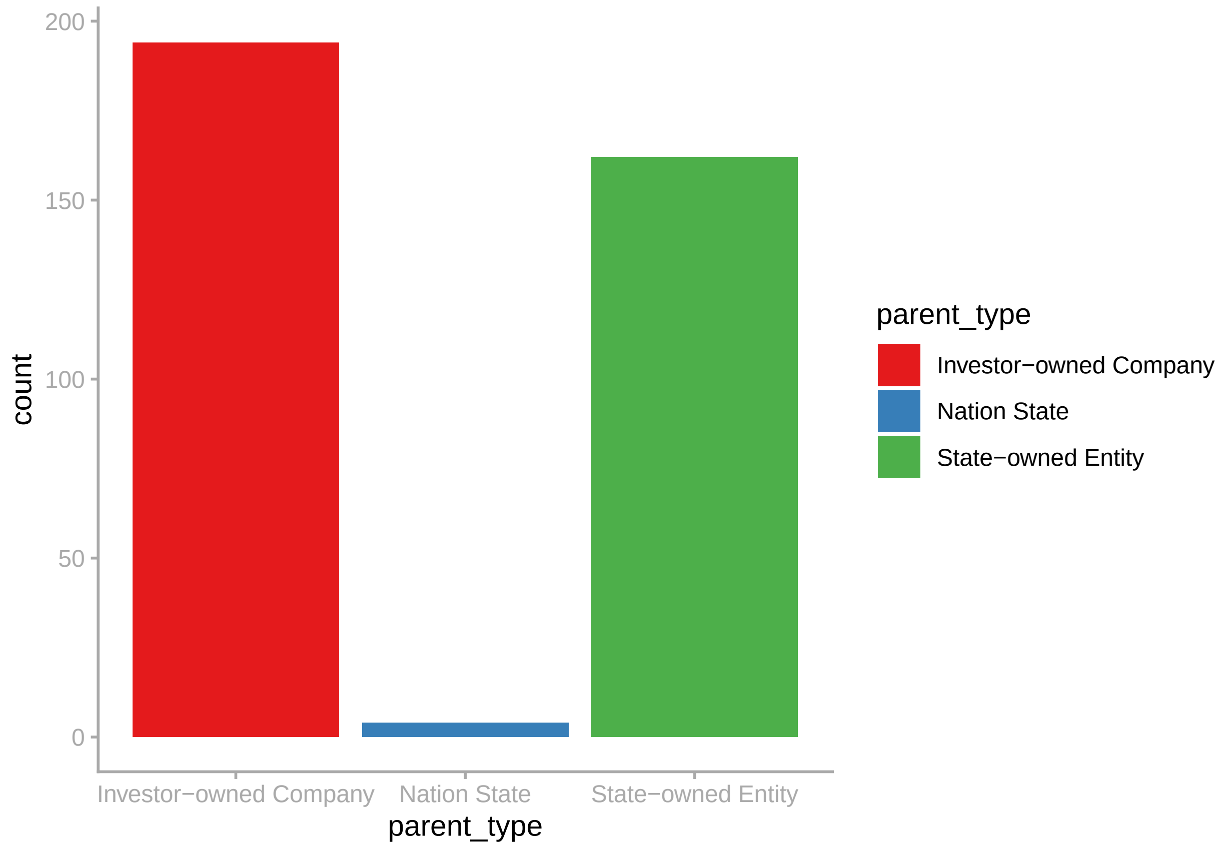
```
library(colorBlindness)
library(ggplot2)
ggplot(emissions_data_24, aes(x=commodity,fill=commodity))+geom_bar()+coord_flip()+ theme_classic()+ the
        axis.ticks = element_line(colour = "darkgrey"),
        axis.text = element_text(colour = "darkgrey"))+
scale_fill_brewer(palette = "Set1")
```




```
table(emissions_data_24$parent_type)
```

```
##
## Investor-owned Company           Nation State     State-owned Entity
##                    194                      4                    162
```

```r
ggplot(emissions_data_24, aes(x=parent_type,fill=parent_type))+geom_bar()+theme_classic()+
   theme(axis.line = element_line(colour = "darkgrey"),
        axis.ticks = element_line(colour = "darkgrey"),
        axis.text = element_text(colour = "darkgrey"))+
  scale_fill_brewer(palette = "Set1")
```




```r
summary(emissions_data_24$"total_emissions_MtCO2e")
```

```
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
## 7.976e-02 1.277e+01 3.874e+01 9.627e+01 1.094e+02 1.498e+03
```

```r
summary(emissions_data_24$"fugitive_methane_emissions_MtCH4")
```

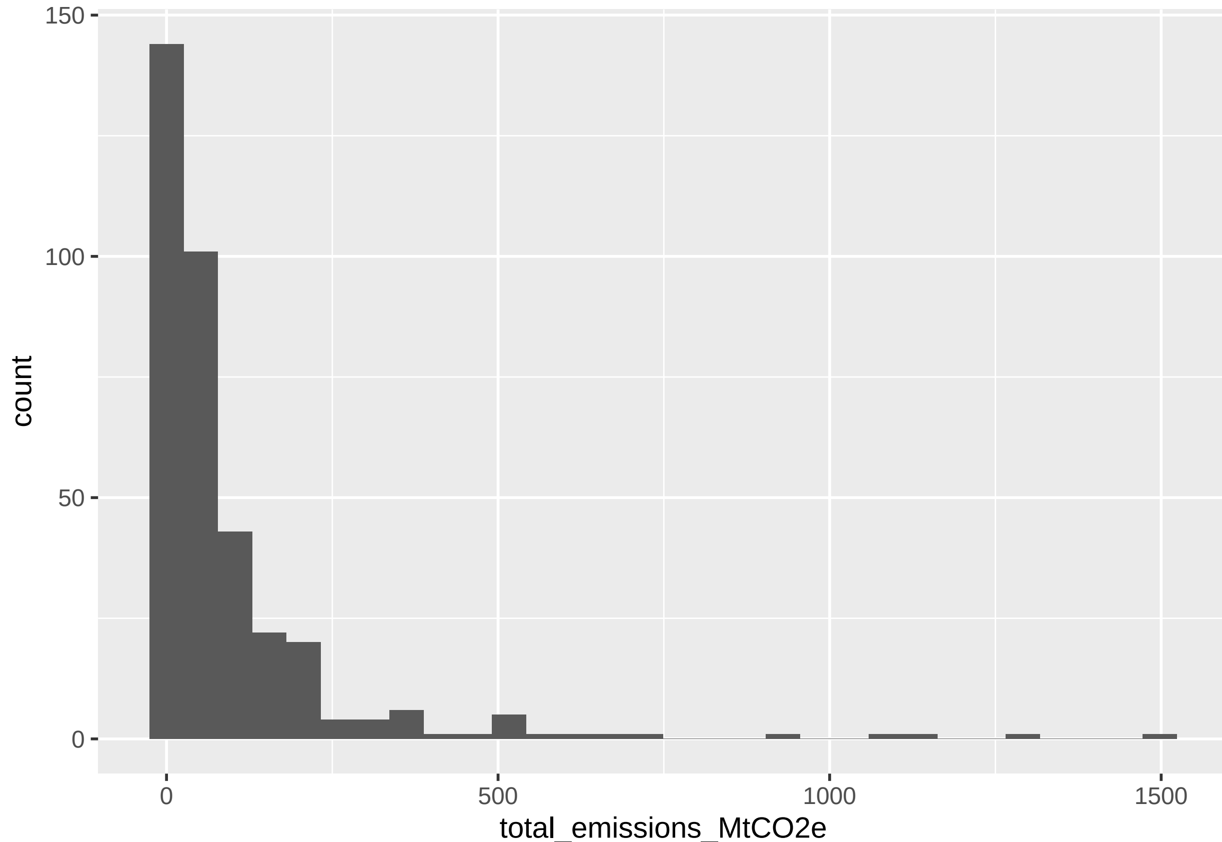
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.03604 0.14343 0.36559 0.39948 7.75792
```

```r
summary(emissions_data_24$"own_fuel_use_emissions_MtCO2")
```

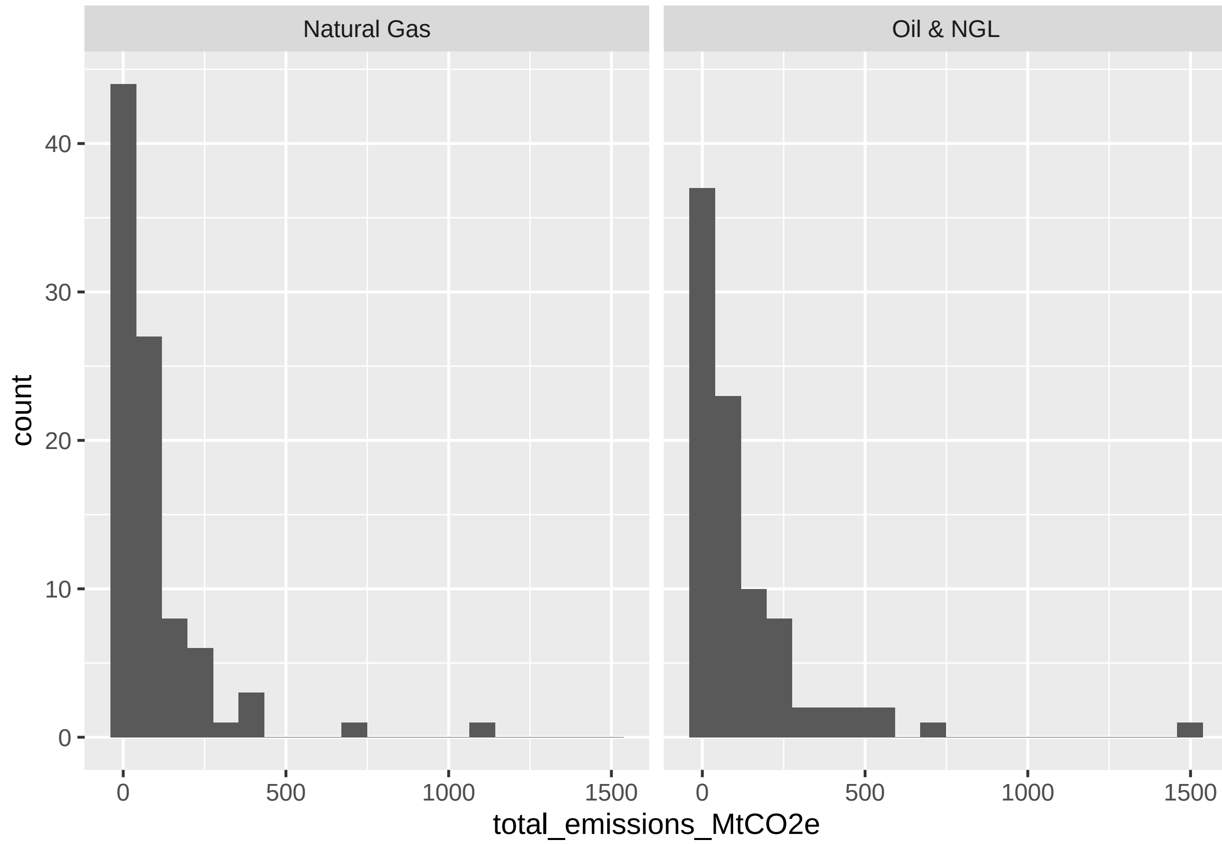
```
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
##  0.000000  0.000000  0.000000  0.974431  0.007591 44.972490
```

```r
ggplot(emissions_data_24, aes(x=total_emissions_MtCO2e))+geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

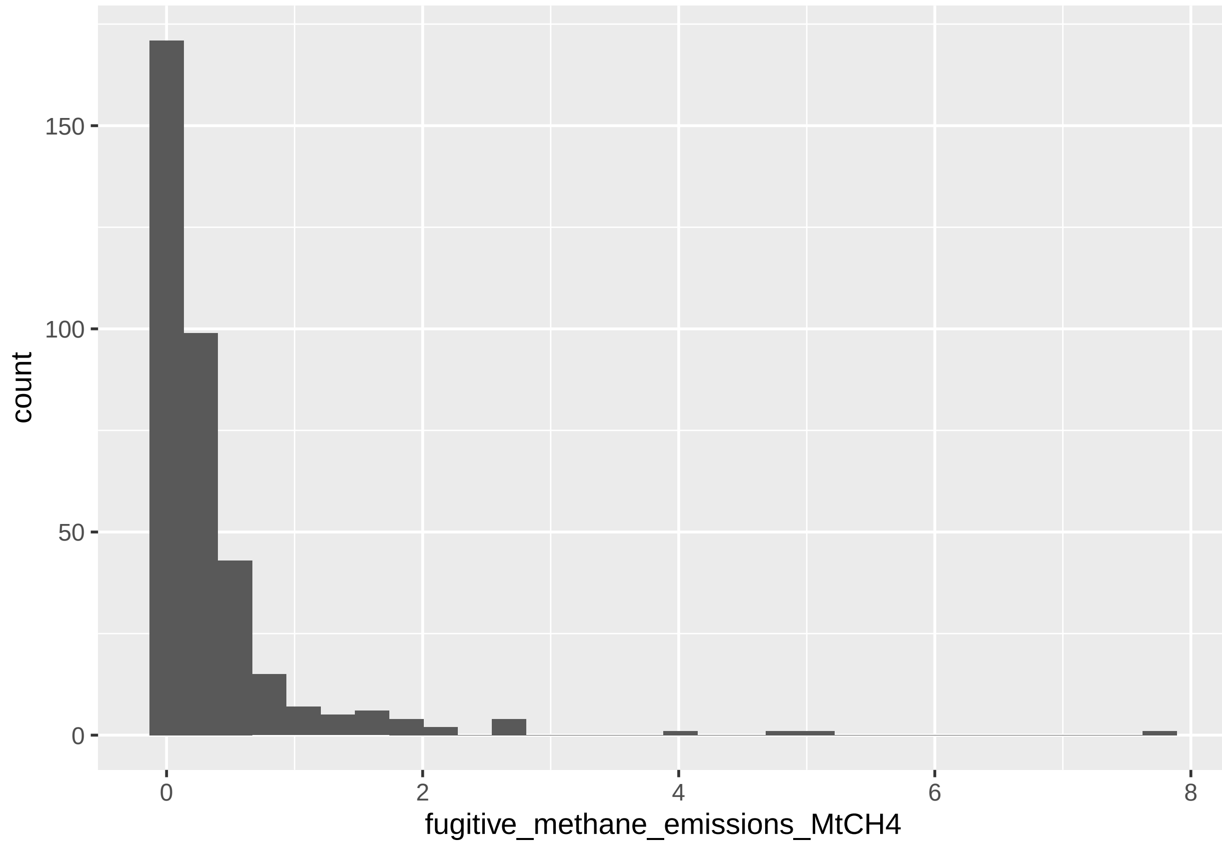



```r
ggplot(data_ng_oil, aes(x = total_emissions_MtCO2e)) +
  geom_histogram( bins =20) +
 facet_grid(~commodity)
```

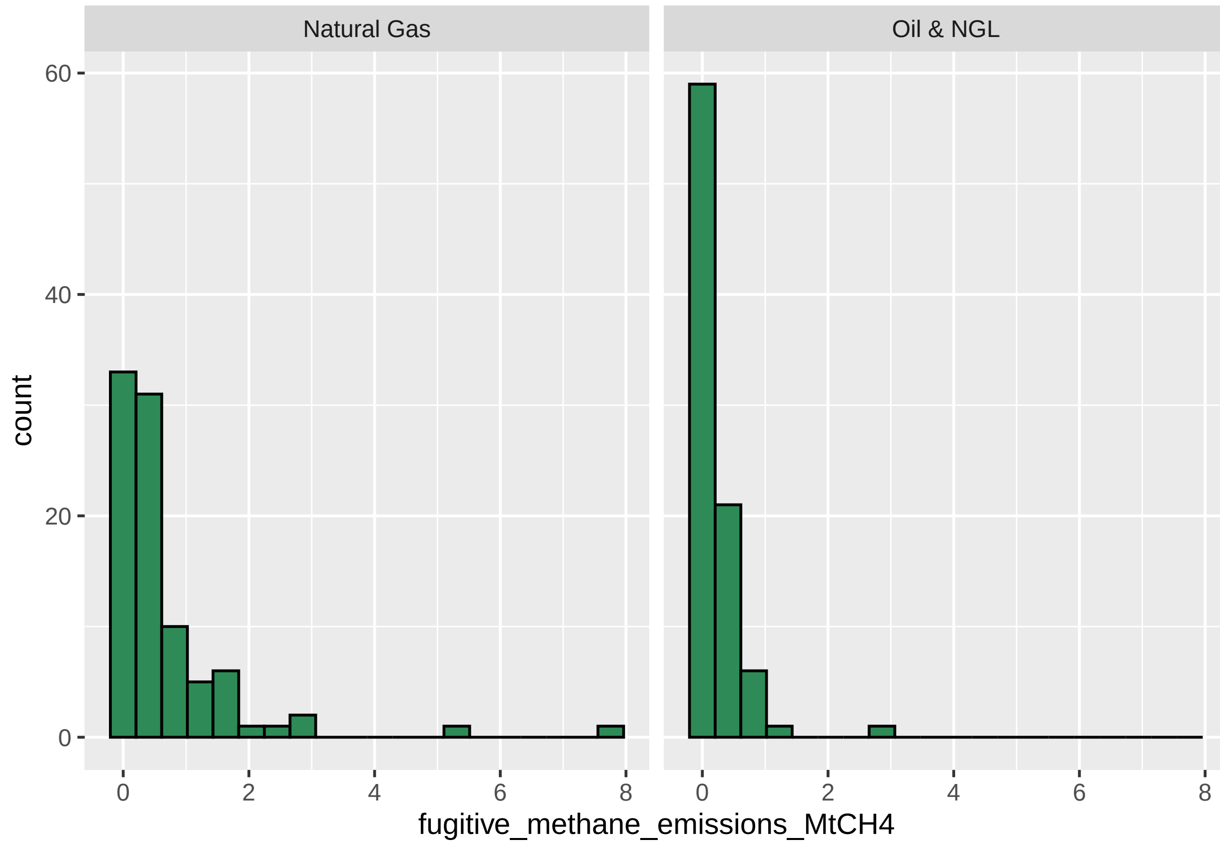



```r
ggplot(emissions_data_24, aes(x=fugitive_methane_emissions_MtCH4))+geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

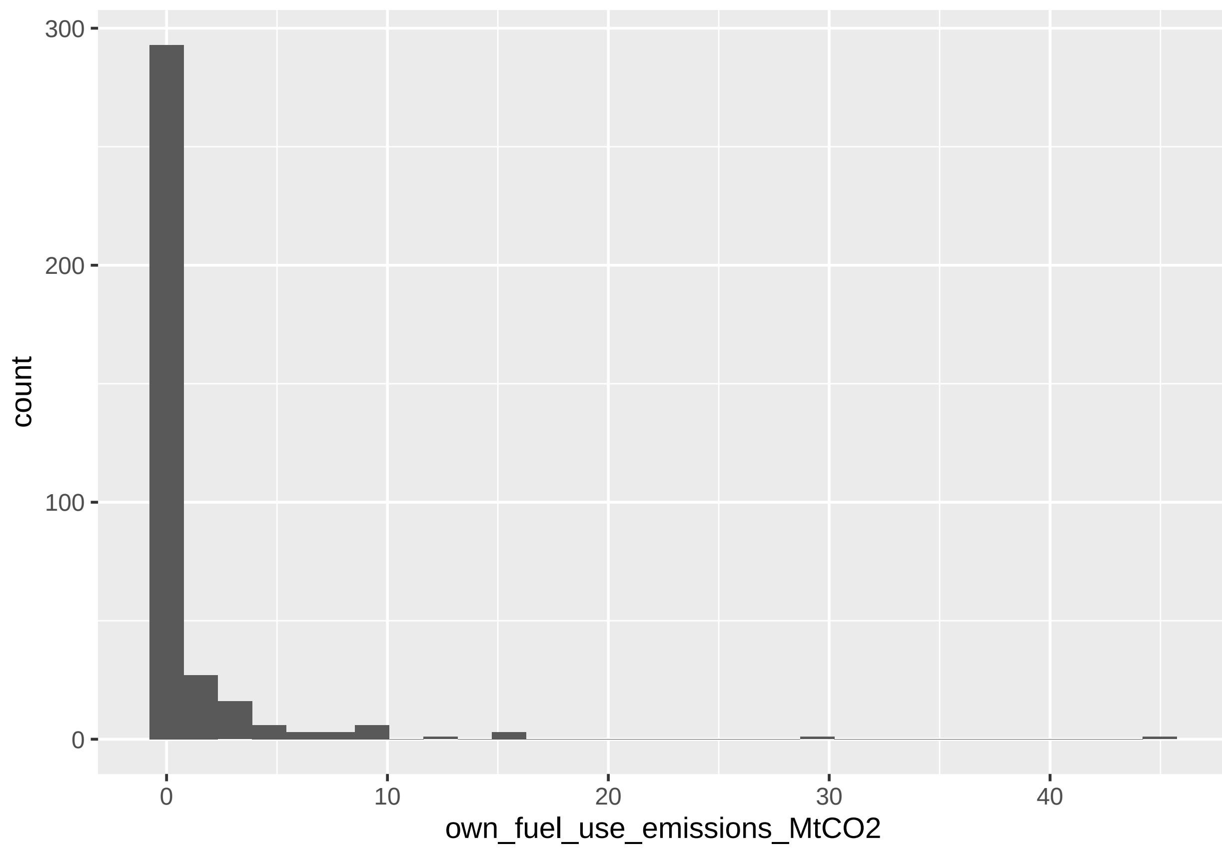



```r
ggplot(data_ng_oil, aes(x = fugitive_methane_emissions_MtCH4)) +
  geom_histogram( bins =20,fill= "seagreen",col="black") +
 facet_grid(~commodity)
```



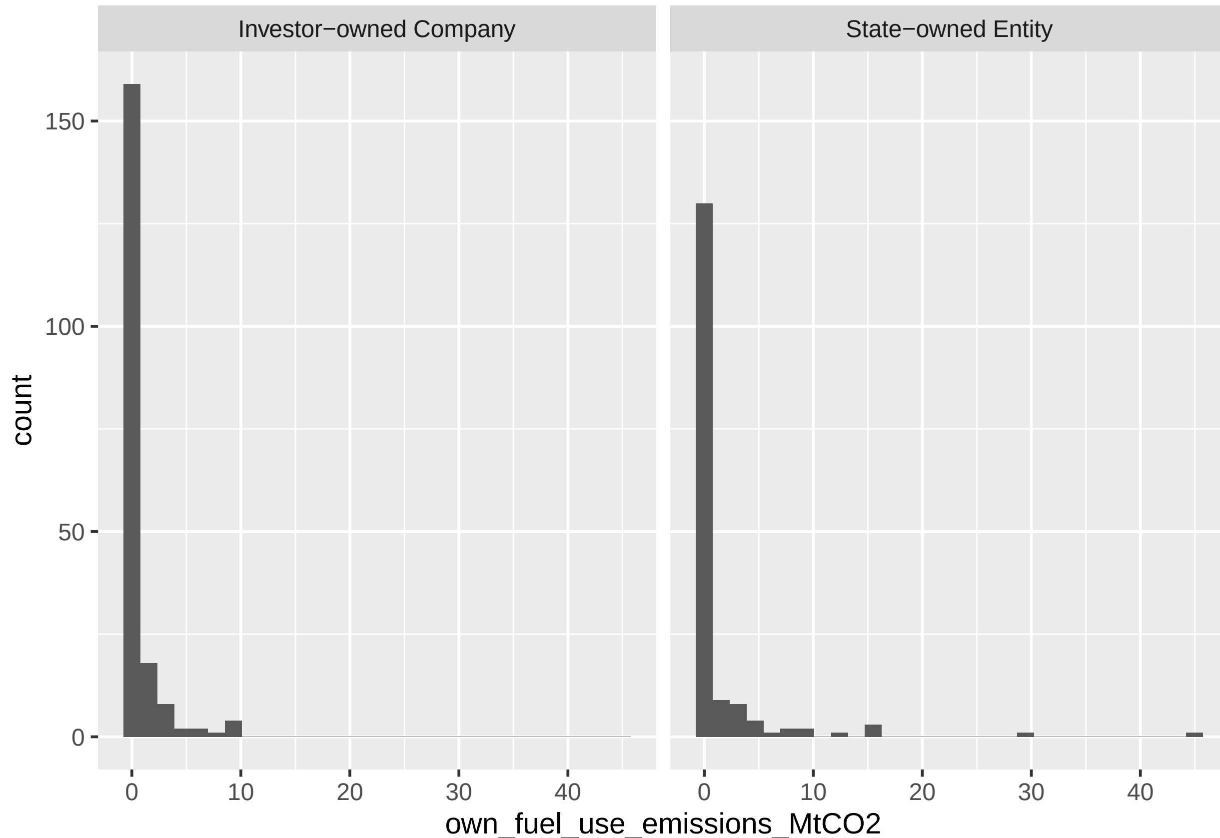

```r
ggplot(emissions_data_24, aes(x=own_fuel_use_emissions_MtCO2))+geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
state_nation=emissions_data_24 %>%
  filter(parent_type %in% c("Investor-owned Company", "State-owned Entity"))

ggplot(state_nation, aes(x=own_fuel_use_emissions_MtCO2))+geom_histogram()+
  facet_grid(~parent_type)
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



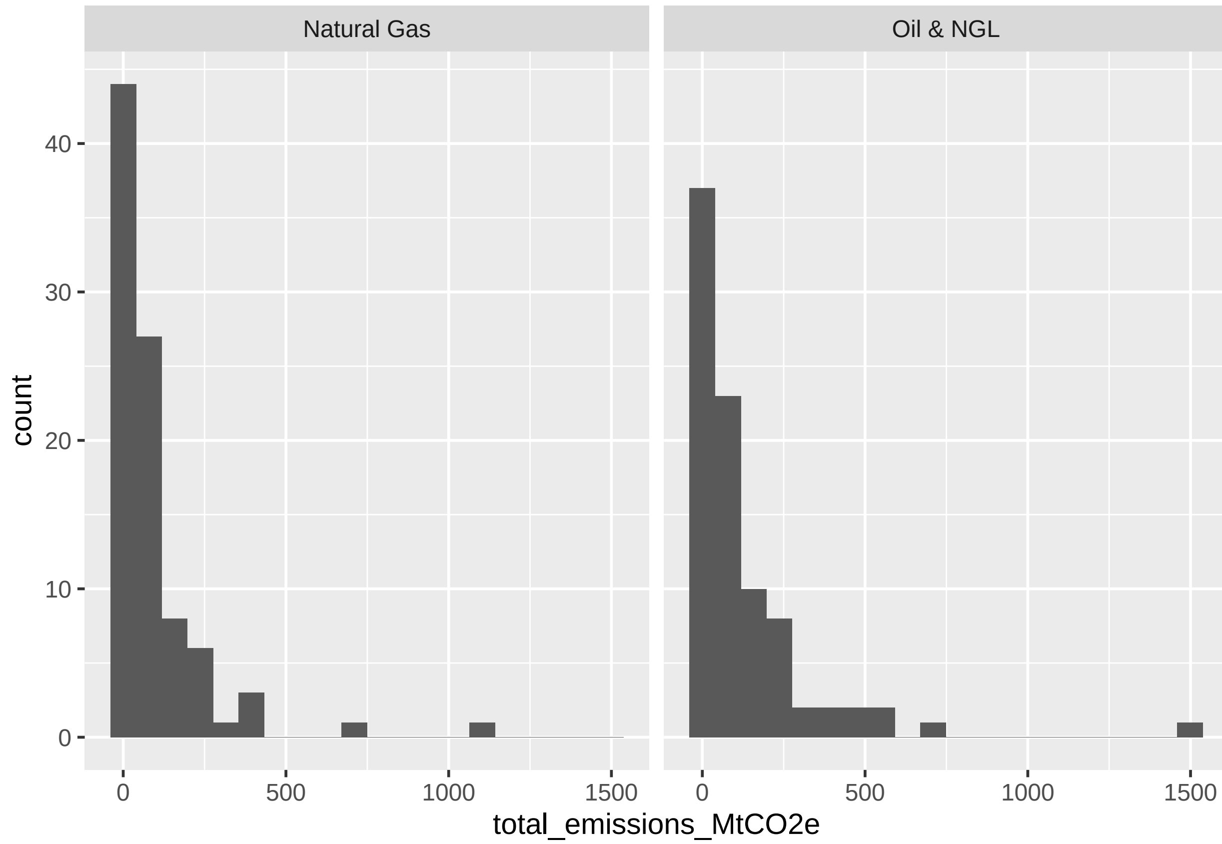
1.Are the mean total emissions of Natural Gas plants significantly different from Oil & NGL plants?

Let μ_1 be the true mean total emissions of Natural Gas plants Let μ_2 be the true mean total emissions of Oil & NGL plants $H_0 : \mu_1 = \mu_2$ $H_a : \mu_1 \neq \mu_2$

```
xbar_1 <- emissions_data_24 %>%
  filter(commodity == "Natural Gas") %>%
  pull(total_emissions_MtCO2e) %>%
  mean()

# A pipline that finds the sample mean of group 2
xbar_2 <- emissions_data_24 %>%
  filter(commodity == "Oil & NGL") %>%
  pull(total_emissions_MtCO2e) %>%
  mean()


# The observed difference in sample means
xbar_diff <- xbar_1 - xbar_2
xbar_diff
```

```
## [1] -32.56679
```

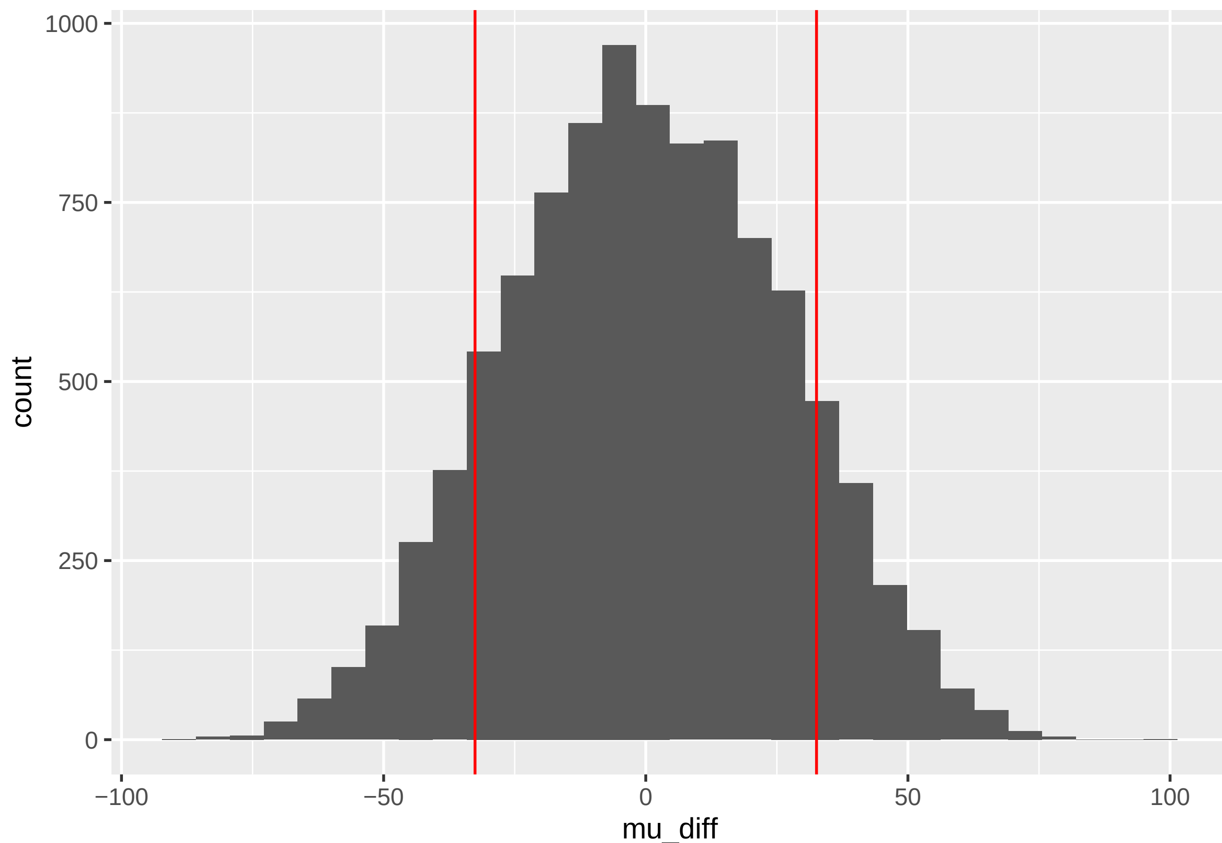
```
ggplot(data_ng_oil, aes(x = total_emissions_MtCO2e)) +
  geom_histogram( bins =20) +
 facet_grid(~commodity)
```




```
set.seed(20)
mu_diff <- replicate(10000, {

  # generate new randomly permuted age groups
  data_ng_oil$permuted_commodity <- sample(data_ng_oil$commodity, replace = FALSE)

  # recompute the difference in means
  mu_1 <- data_ng_oil %>%
    filter(permuted_commodity == "Natural Gas") %>%
    pull(total_emissions_MtCO2e) %>%
    mean()
  mu_2 <- data_ng_oil %>%
    filter(permuted_commodity == "Oil & NGL") %>%
    pull(total_emissions_MtCO2e) %>%
    mean()
  mu_1 - mu_2
})
```

```
data.frame(mu_diff) %>% ggplot(aes(x = mu_diff)) +
  geom_histogram() +
  geom_vline(xintercept = xbar_diff, color = "red")+
  geom_vline(xintercept = -xbar_diff, color = "red")
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
pvalue <- mean(mu_diff > abs(xbar_diff)) + mean(mu_diff < -abs(xbar_diff))
pvalue
```

```
## [1] 0.2286
```

There is little evidence to support rejecting the null hypothesis or conclude that the total emissions from Natural Gas Plants are different than Oil & NG Plants.

Question 2. Test whether the mean fugitive methane emissions of Natural Gas plants differ significantly from Oil & NGL plants.

H0: mu fugitive methane emissions of Natural Gas plants = mu fugitive methane emissions of Oil & NGL plants.

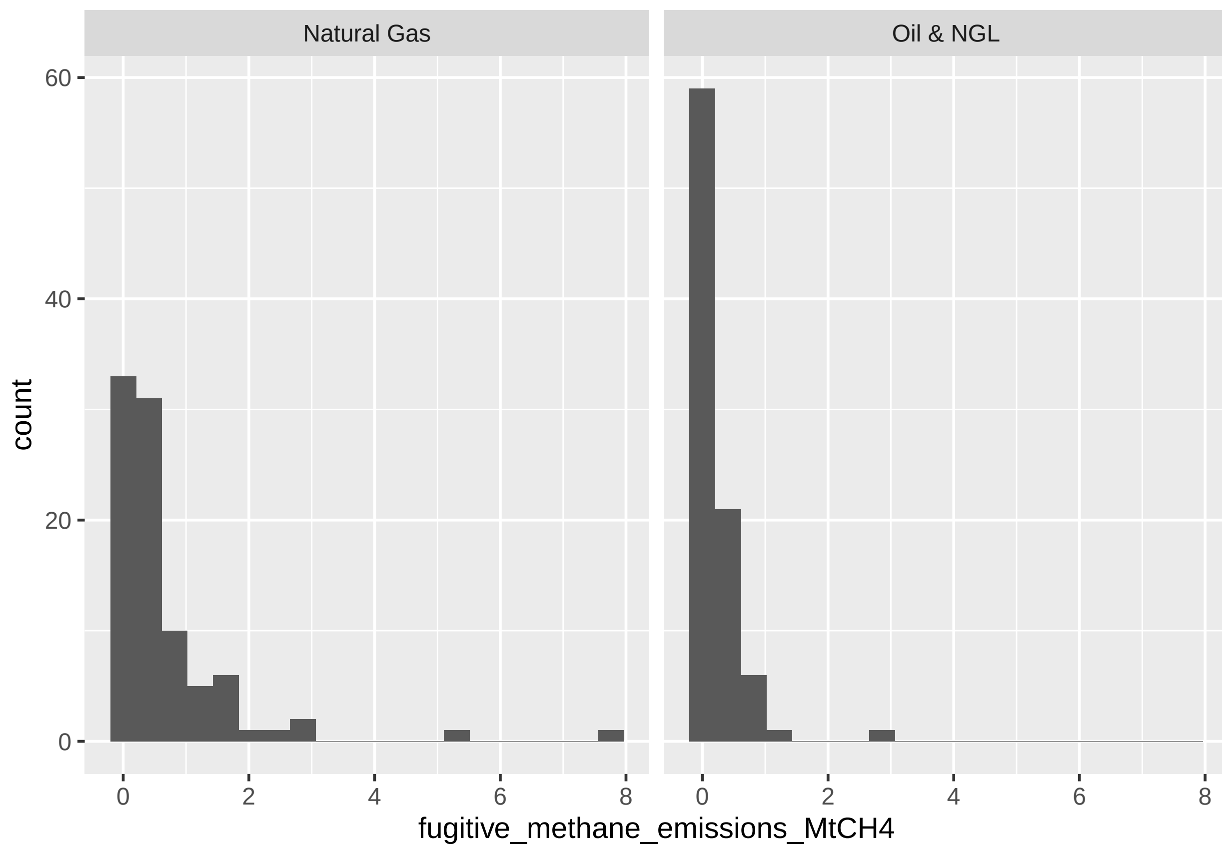
HA: mu fugitive methane emissions of Natural Gas plants =/ mu fugitive methane emissions of Oil & NGL plants.

```
xbar_natural_gas <- emissions_data_24 %>%
  filter(commodity == "Natural Gas") %>%
  pull(fugitive_methane_emissions_MtCH4) %>%
  mean()

xbar_oil_ngl <- emissions_data_24 %>%
  filter(commodity == "Oil & NGL") %>%
  pull(fugitive_methane_emissions_MtCH4) %>%
  mean()
```

```r
#The difference in means
xbar_diff <- xbar_natural_gas - xbar_oil_ngl
xbar_diff
```

```
## [1] 0.4421161
```

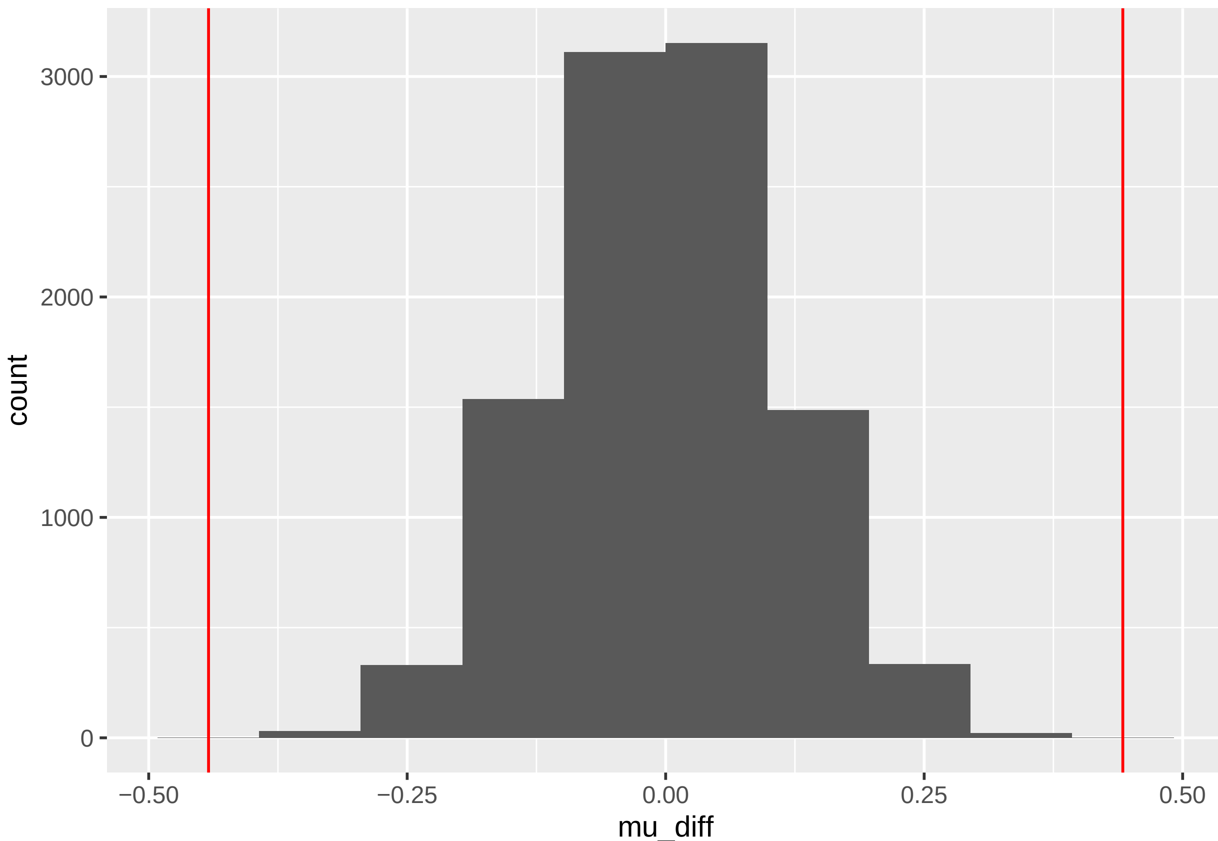
```r
plot_data<-emissions_data_24 %>% filter(commodity == "Natural Gas" | commodity == "Oil & NGL")
ggplot(plot_data, aes(x = fugitive_methane_emissions_MtCH4)) +
  geom_histogram( bins =20) +
  facet_grid(~commodity)
```




```r
set.seed(20)
mu_diff <- replicate(10000, {
  # generate new randomly permuted age groups
  emissions_data_24$permuted <- sample(emissions_data_24$commodity)

  # recompute the difference in means
  mu_A <- emissions_data_24 %>%
    filter(permuted == "Natural Gas") %>%
    pull(fugitive_methane_emissions_MtCH4) %>%
    mean()
  mu_B <- emissions_data_24 %>%
    filter(permuted == "Oil & NGL") %>%
    pull(fugitive_methane_emissions_MtCH4) %>%
```

```r
    mean()
  mu_A - mu_B
})
```

```r
data.frame(mu_diff) %>% ggplot(aes(x = mu_diff)) +
  geom_histogram(bins = 10) +
  geom_vline(xintercept = xbar_diff, color = "red")+
  geom_vline(xintercept = -xbar_diff, color = "red")
```




```r
2 * mean(mu_diff > xbar_diff)
```

```
## [1] 0
```

Strong evidence against the null, fugitive methane emissions of Natural Gas plants is significantly different from fugitive methane emissions of Oil & NGL plants.

Question 3. Using a permutation test, determine whether the mean own fuel use emissions differs significantly between state-owned and investor-owned

H0: mu own fuel use emissions state-owned = mu own fuel use emissions investor-owned

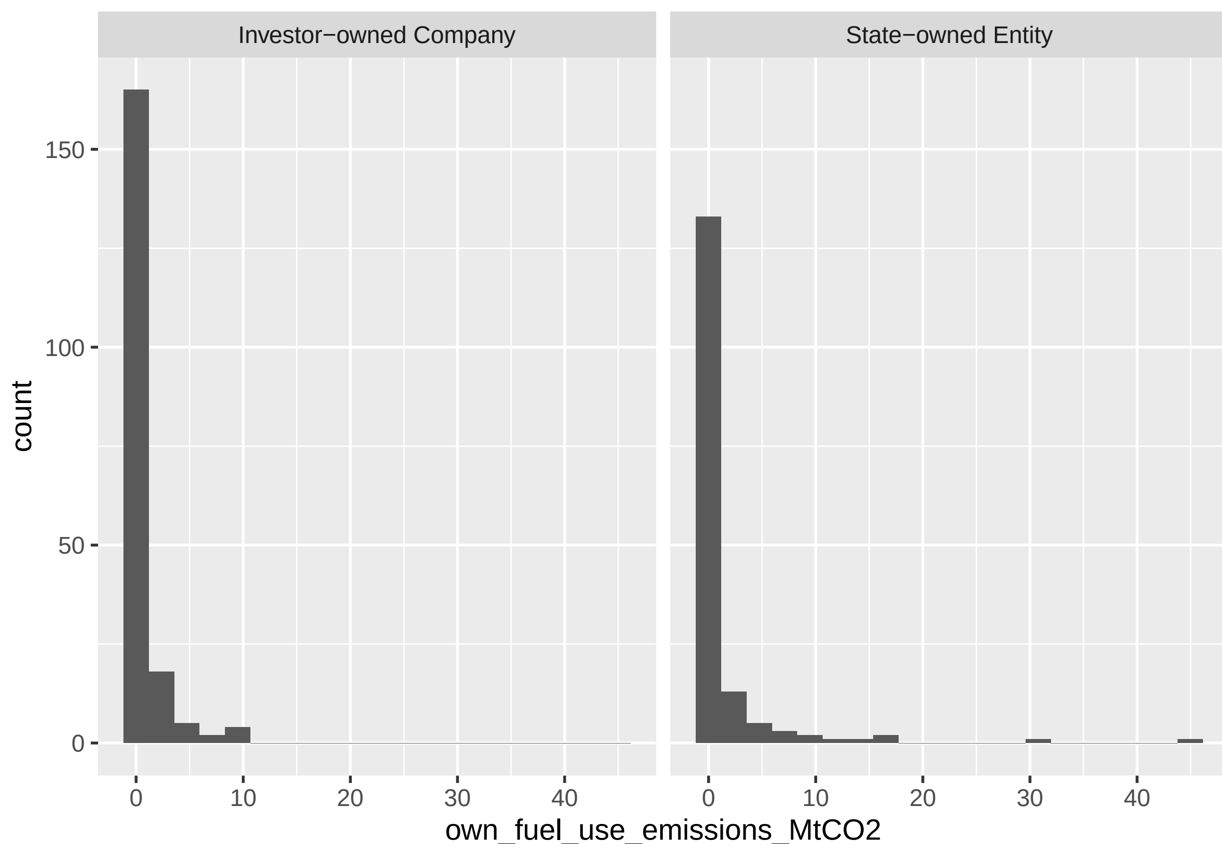
HA: mu own fuel use emissions state-owned =/ mu own fuel use emissions investor-owned

```r
xbar_1 <- emissions_data_24 %>%
  filter(parent_type == "State-owned Entity") %>%
```

```r
  pull(own_fuel_use_emissions_MtCO2) %>%
  mean()

xbar_2 <- emissions_data_24 %>%
  filter(parent_type == "Investor-owned Company") %>%
  pull(own_fuel_use_emissions_MtCO2) %>%
  mean()


# The observed difference in sample means
xbar_diff <- xbar_1 - xbar_2
xbar_diff
```

```
## [1] 0.8181274
```

```r
plot_data2 <- emissions_data_24%>% filter(parent_type == "Investor-owned Company" | parent_type == "Sta
ggplot(plot_data2, aes(x = own_fuel_use_emissions_MtCO2)) +
  geom_histogram( bins =20) +
 facet_grid(~parent_type)
```

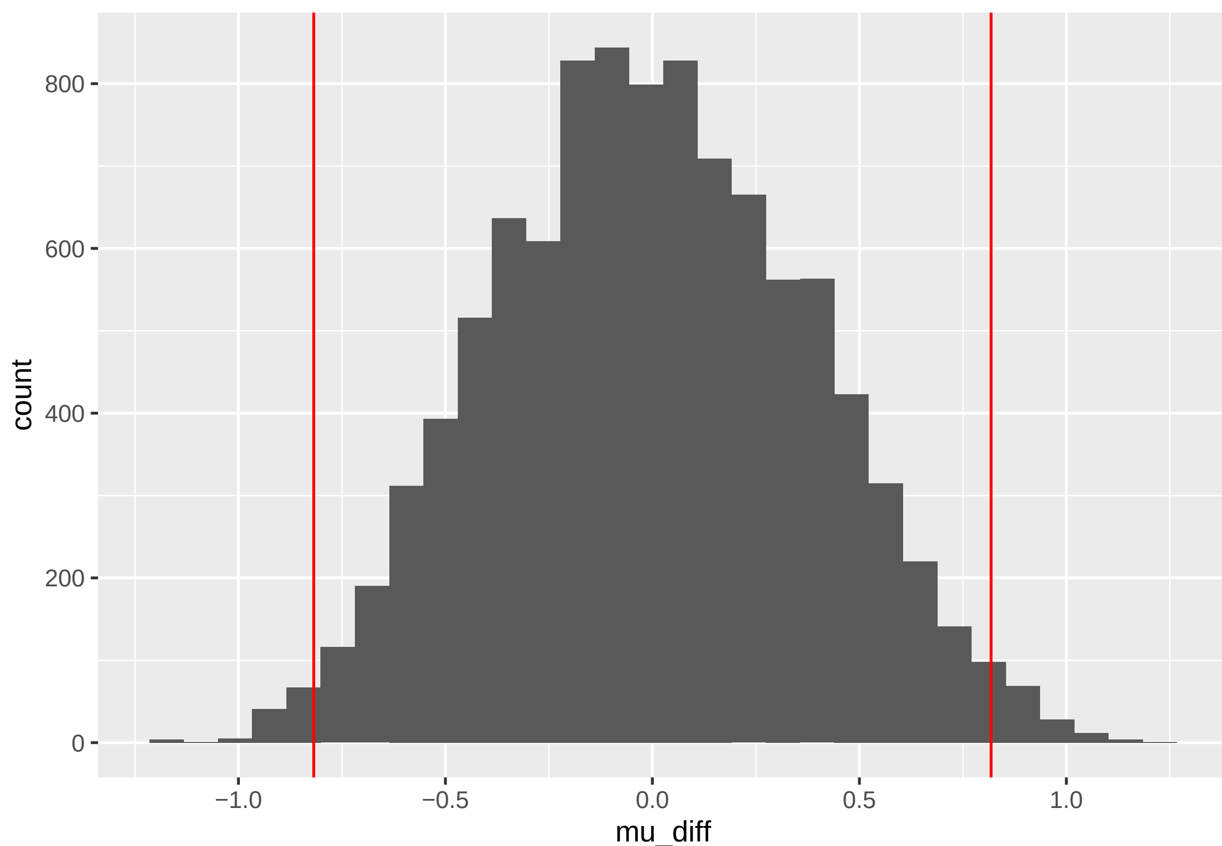



```r
set.seed(20)
mu_diff <- replicate(10000, {

  emissions_data_24$permuted_parent <- sample(emissions_data_24$parent_type, replace = FALSE)
```

```
  # recompute the difference in means
  mu_1 <- emissions_data_24 %>%
    filter(permuted_parent == "State-owned Entity") %>%
    pull(own_fuel_use_emissions_MtCO2) %>%
    mean()
  mu_2 <- emissions_data_24 %>%
    filter(permuted_parent == "Investor-owned Company") %>%
    pull(own_fuel_use_emissions_MtCO2) %>%
    mean()
  mu_1 - mu_2
})
```

```
data.frame(mu_diff) %>% ggplot(aes(x = mu_diff)) +
  geom_histogram() +
  geom_vline(xintercept = xbar_diff, color = "red")+
  geom_vline(xintercept = -xbar_diff, color = "red")
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
pvalue <- mean(mu_diff > abs(xbar_diff)) + mean(mu_diff < -abs(xbar_diff))
pvalue
```

```
## [1] 0.0244
```

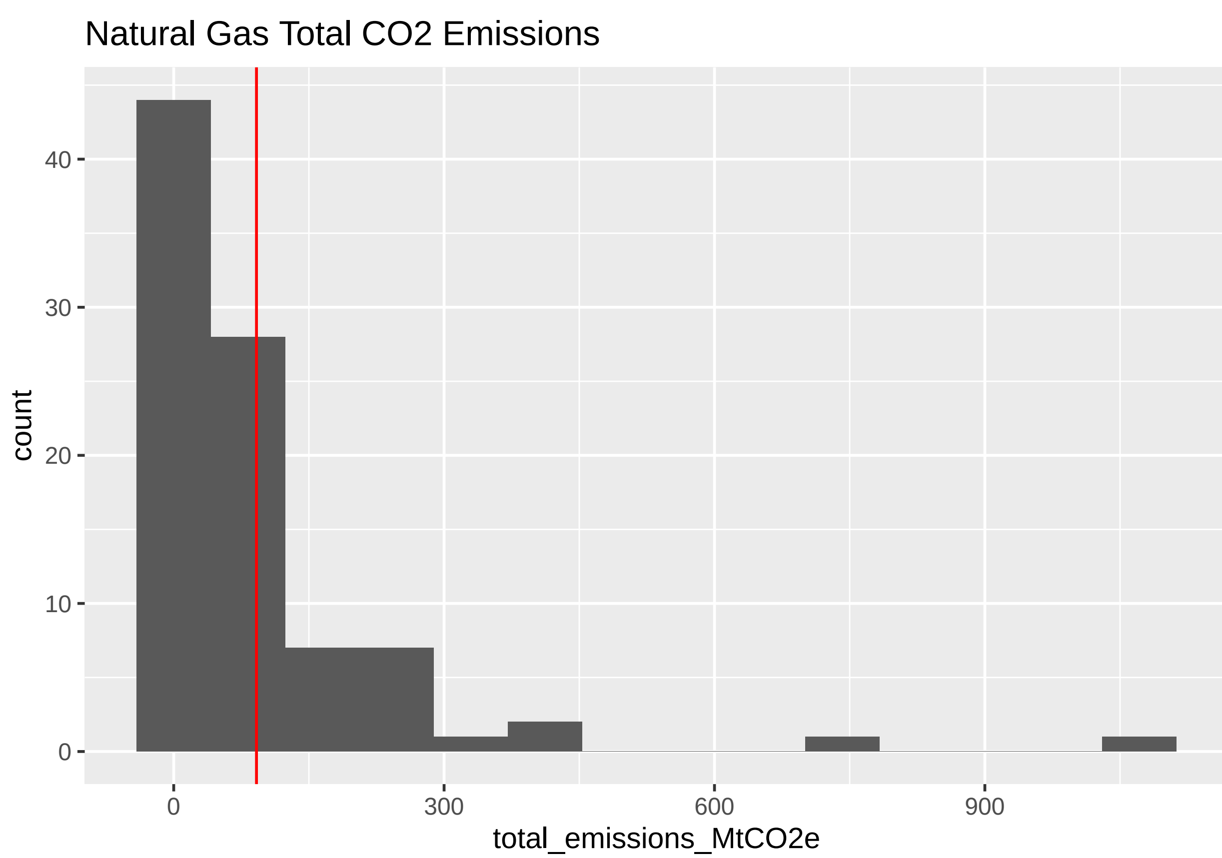
Strong evidence against the null, state-owned mean own fuel use emissions are statistically different from investor-owned mean own fuel use emissions.

Question 4. 95% confidence interval for the mean total emissions of Natural Gas plants

```
nat_gas_data <- emissions_data_24%>%filter(commodity == "Natural Gas")

xbar <- nat_gas_data %>%
  pull(total_emissions_MtCO2e) %>%
  mean()

xbar
```

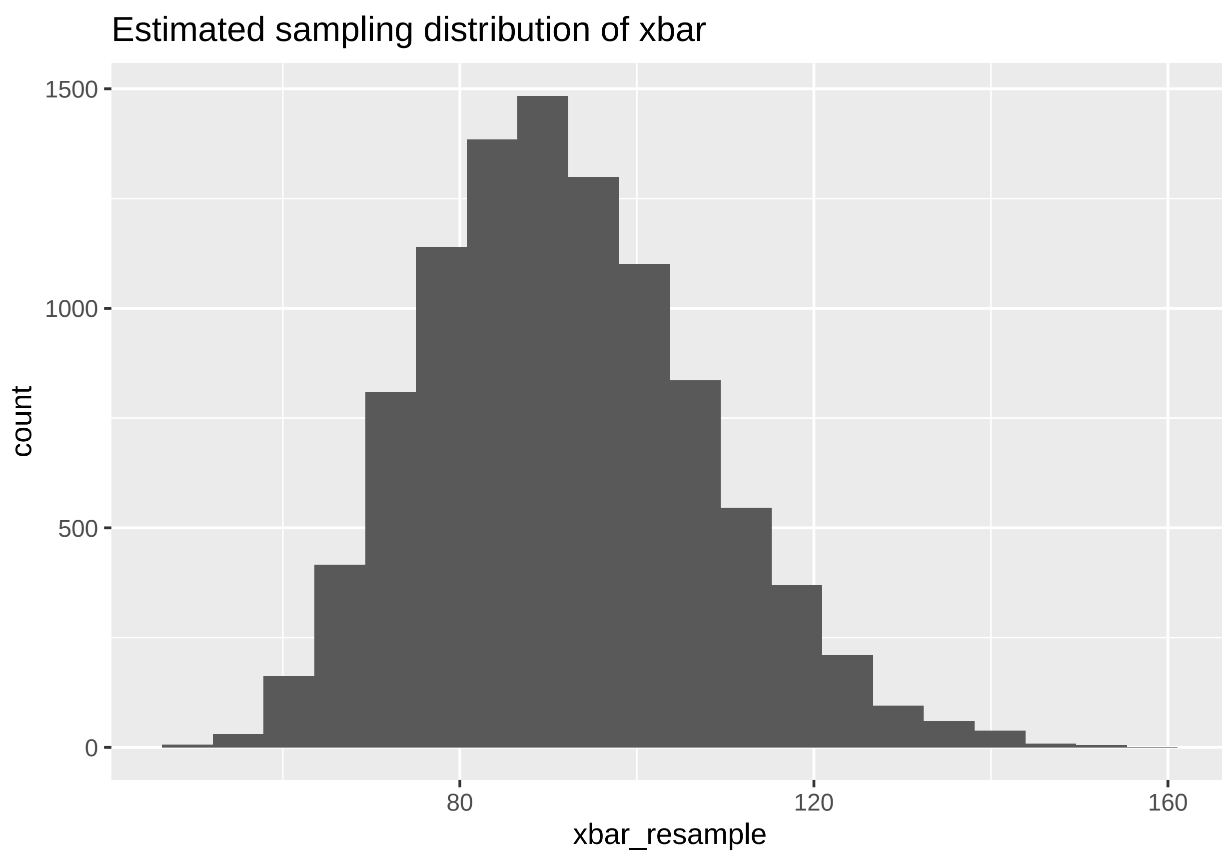
```
## [1] 91.82953
```

```
ggplot(nat_gas_data, aes(x = total_emissions_MtCO2e)) +
  geom_histogram(bins = 14) +
  geom_vline(xintercept = xbar, color = "red")+
  ggtitle("Natural Gas Total CO2 Emissions")
```




```
set.seed(20)
xbar_resample  <- replicate(10000, {
  tot_co2_emissions_resample <- sample(nat_gas_data$total_emissions_MtCO2e, replace = TRUE)
  mean(tot_co2_emissions_resample)
})
```

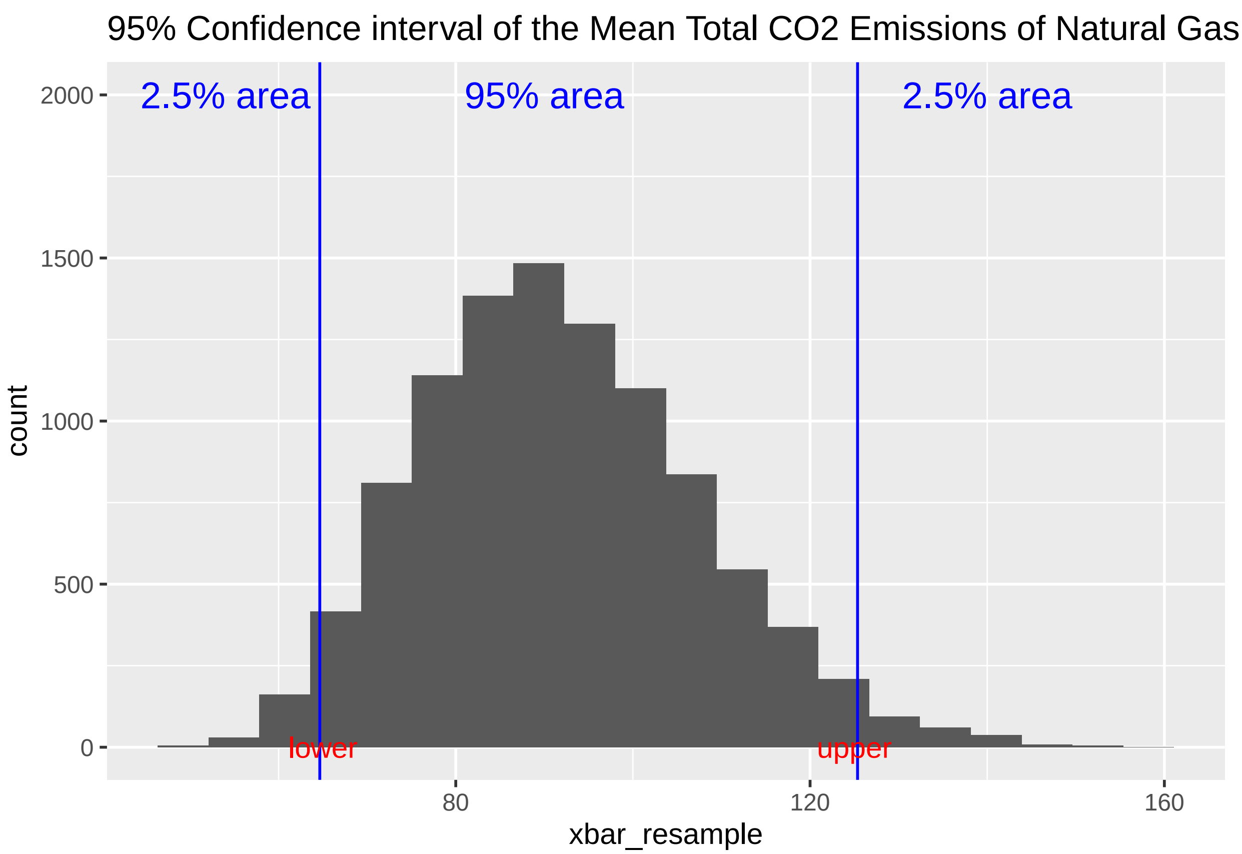
```r
ggplot(data.frame(xbar_resample), aes(x = xbar_resample)) +
  geom_histogram(bins=20) +
  ggtitle("Estimated sampling distribution of xbar")
```




```r
set.seed(20)
quantile(xbar_resample, c(0.025, 0.975))
```

```
##      2.5%     97.5%
##  64.65603 125.36208
```

```r
ggplot(data.frame(xbar_resample), aes(x = xbar_resample)) +
  geom_histogram(bins=20) +
  geom_vline(xintercept = c(64.65, 125.36), col = "blue")+
  ggtitle("95% Confidence interval of the Mean Total CO2 Emissions of Natural Gas Plants")+
  annotate("text", x=90, y=2000, label= "95% area", color = 'blue', size =5)+
  annotate("text", x=54, y=2000, label= "2.5% area", color = 'blue', size =5)+
  annotate("text", x=140, y=2000, label= "2.5% area", color = 'blue', size =5)+
  annotate("text", x=125, y=0, label= "upper", color = 'red', size =4)+
  annotate("text", x=65, y=0, label= "lower", color = 'red', size =4)
```



The confidence interval calculated allows us to estimate the mean total CO2 emissions of Natural Gas plants

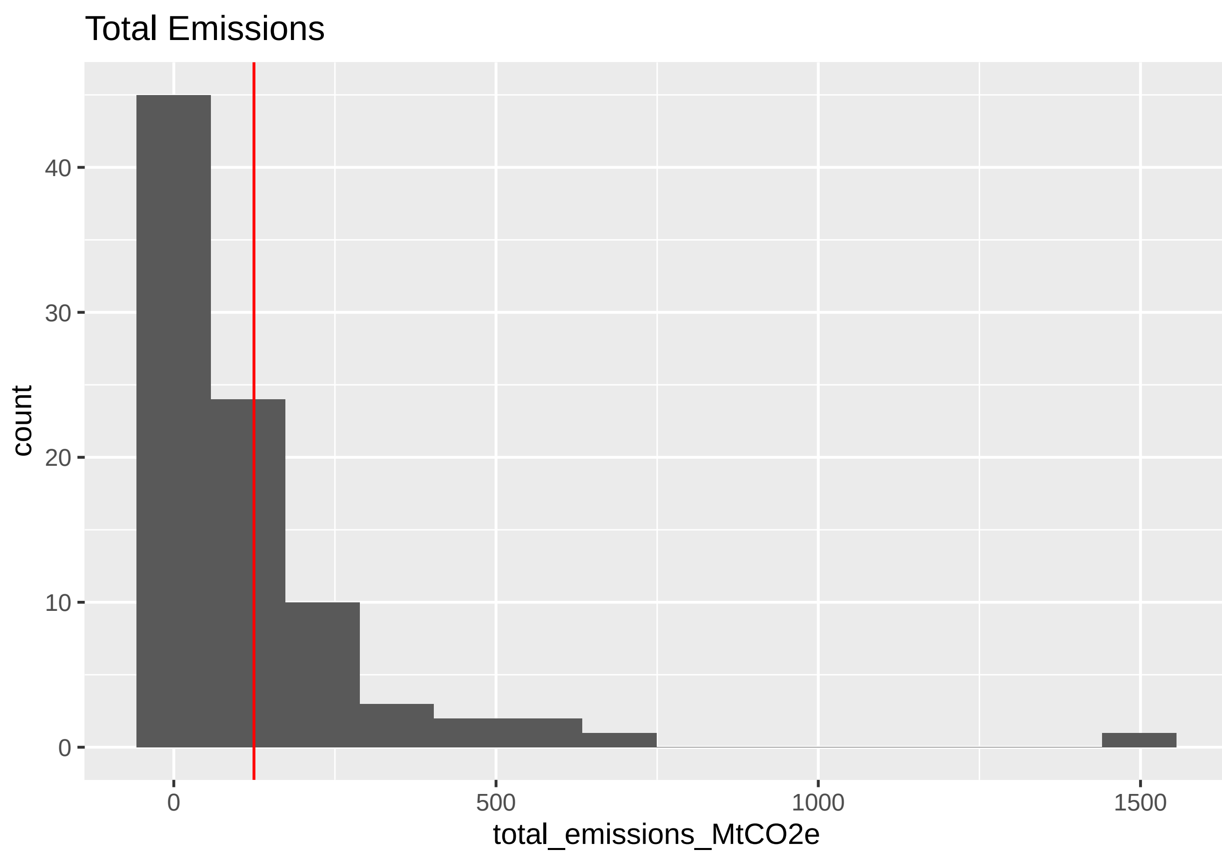
Specifically, we are 95% confident that the mean total CO2 emissions of Natural Gas plants is between 64.65603 and 125.36208 metric tons.

5.What is the 95% confidence interval for the mean total emissions of Oil & NGL plants?

```
data_ng=emissions_data_24 %>%
  filter(commodity %in% c("Natural Gas"))

data_oil=emissions_data_24 %>%
  filter(commodity %in% c("Oil & NGL"))

# xbar calculation
xbar <- data_oil %>%
  pull(total_emissions_MtCO2e) %>%
  mean()

xbar
```

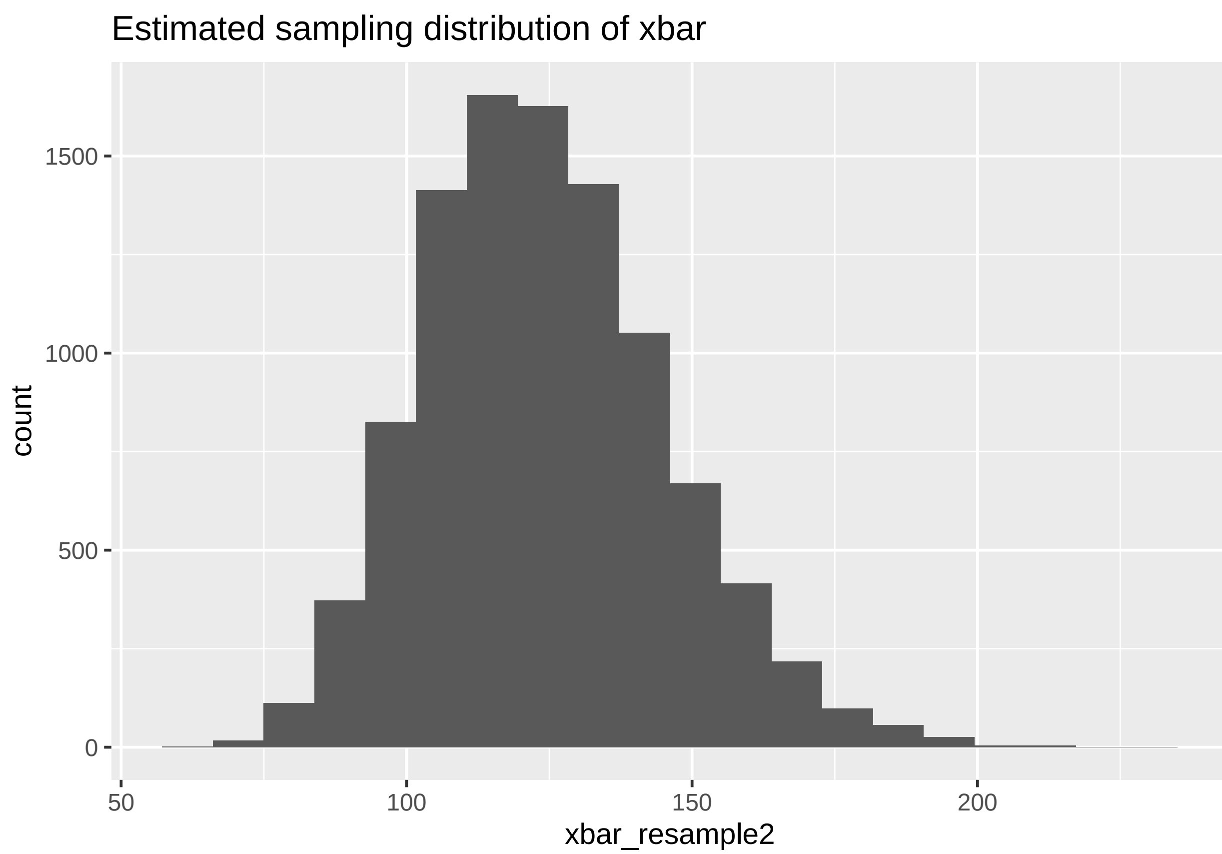
```
## [1] 124.3963
```

```
ggplot(data_oil, aes(x = total_emissions_MtCO2e)) +
  geom_histogram(bins = 14) +
  geom_vline(xintercept = xbar, color = "red")+
  ggtitle("Total Emissions")
```




```r
set.seed(20)
xbar_resample2  <- replicate(10000, {
  total_emission_resample <- sample(data_oil$total_emissions_MtCO2e, replace = TRUE)
  mean(total_emission_resample)
})
```

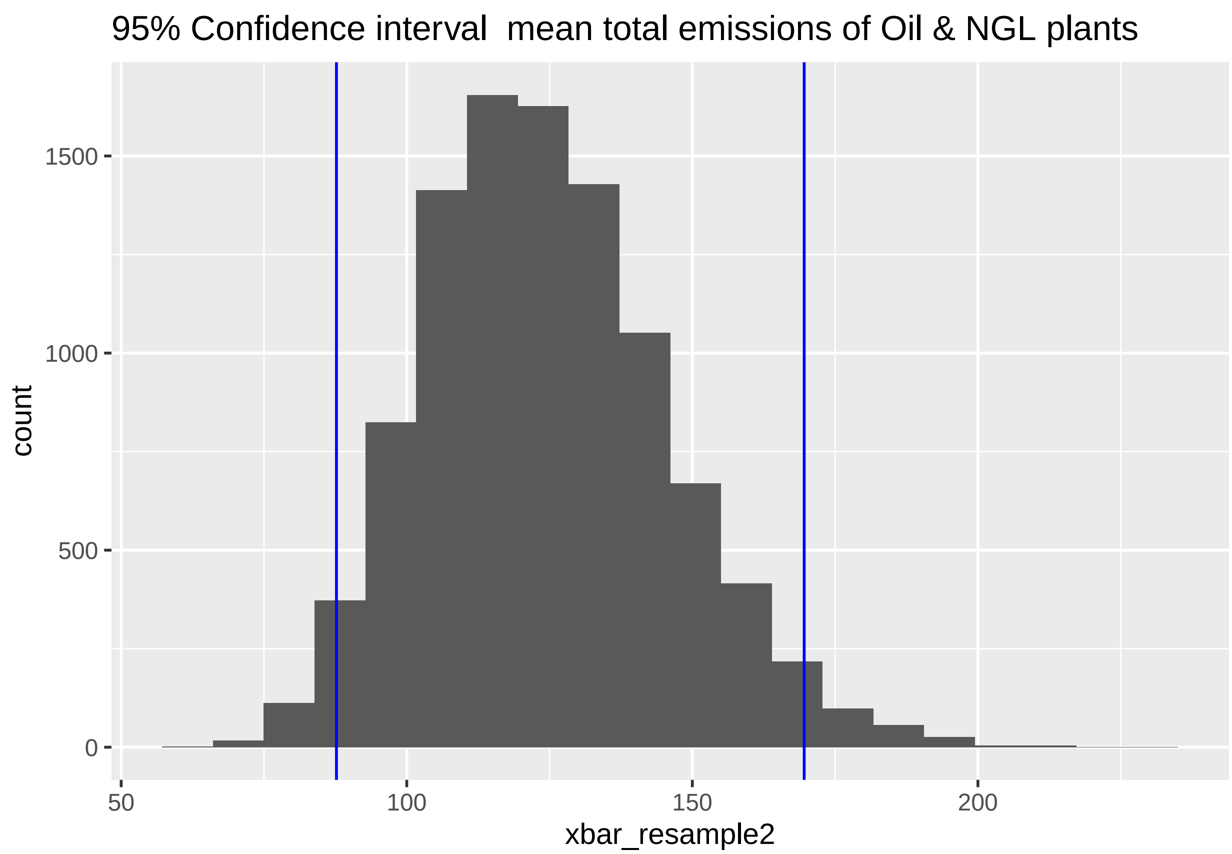
```r
ggplot(data.frame(xbar_resample2), aes(x = xbar_resample2)) +
  geom_histogram(bins=20) +
  ggtitle("Estimated sampling distribution of xbar")
```




```
quantile(xbar_resample2, c(0.025, 0.975))
```

```
##      2.5%     97.5%
##  87.69647 169.57829
```

```
ggplot(data.frame(xbar_resample2), aes(x = xbar_resample2)) +
  geom_histogram(bins=20) +
  geom_vline(xintercept = c(87.69, 169.58), col = "blue")+
  ggtitle("95% Confidence interval  mean total emissions of Oil & NGL plants")
```



The confidence interval calculated allows us to estimate the mean total emissions of Oil & NG plants

Specifically, we are 95% confident that the mean total CO2 emissions of Oil & NG plants is between 87.69647 and 169.57829 metric tons.

6.What is the 95% confidence interval for the difference of mean fugitive methane between Natural Gas Plants and Oil & NGL?

```
set.seed(20)

fugi_data_ng= emissions_data_24%>%
  filter(commodity %in% c("Natural Gas")) %>%
  pull(fugitive_methane_emissions_MtCH4)

xbar_boot_1  <- replicate(10000, {
  fugi_data_ng_resample <- sample(fugi_data_ng,
                                  replace = TRUE)
  mean(fugi_data_ng_resample)
})
```

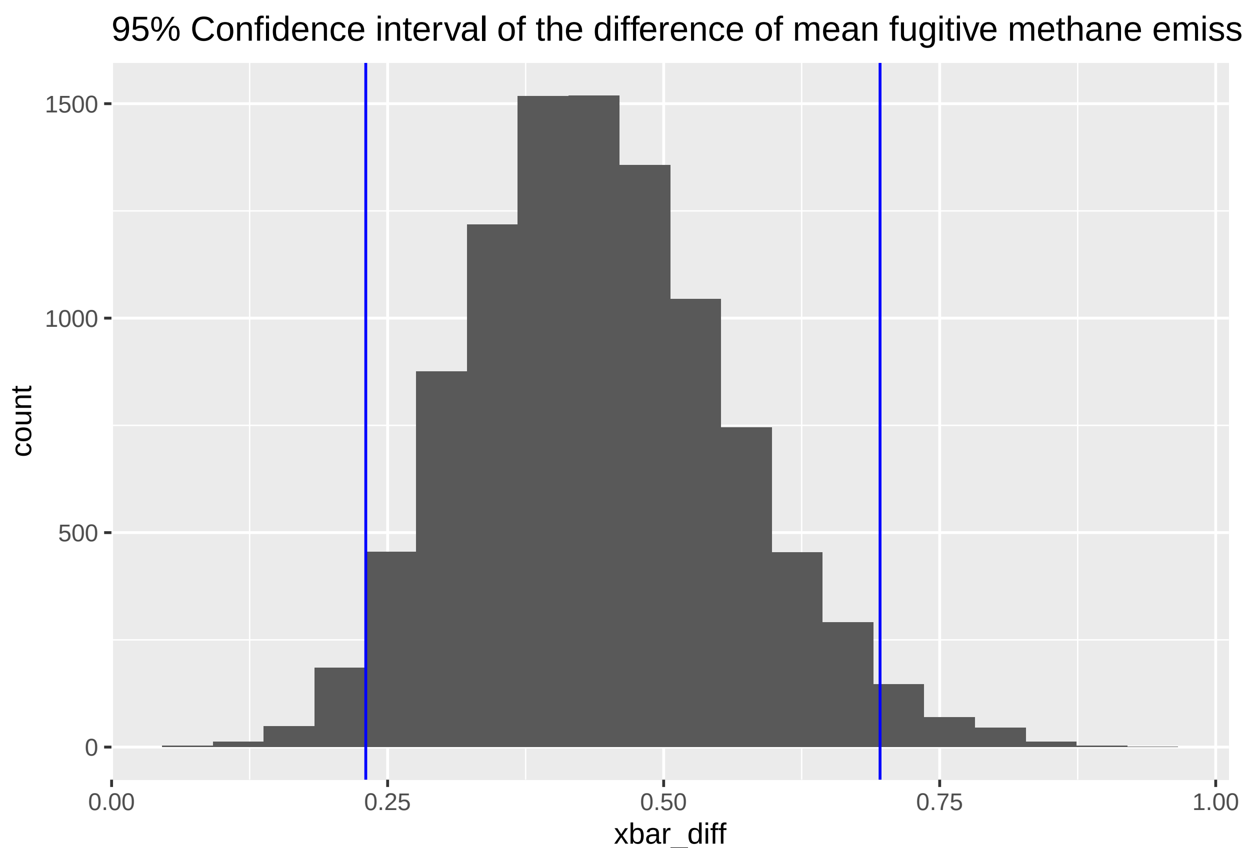
```
set.seed(20)

fugi_data_oil=emissions_data_24 %>%
  filter(commodity %in% c("Oil & NGL")) %>%
  pull(fugitive_methane_emissions_MtCH4)

xbar_boot_2  <- replicate(10000, {
  fugi_data_oil_resample <- sample(fugi_data_oil,
```

```
                                         replace = TRUE)
  mean(fugi_data_oil_resample)
})
```

```
xbar_diff <- xbar_boot_1-xbar_boot_2
quantile(xbar_diff, c(0.025, 0.975))
```

```
##      2.5%     97.5%
## 0.2301867 0.6960284
```

```
ggplot(data.frame(xbar_diff), aes(x = xbar_diff)) +
  geom_histogram(bins=20) +
  geom_vline(xintercept = c(.2302, .6960), col = "blue")+
  ggtitle("95% Confidence interval of the difference of mean fugitive methane emissions")
```



The confidence interval calculated allows us to estimate the mean difference of fugitive methane emissions between natural gas plants and OIl & NG plants.

Specifically, we are 95% confident that the mean difference of fugitive methane emissions is between .2301867 and .6960284 metric tons.

Question 7. Given an entity's total co2 emissions or fugitive methane emissions, predict whether it came from a natural gas or oil & ngl plant.

The confidence interval calculated allows us to estimate the mean difference of fugitive methane emissions between natural gas plants and OIl & NG plants.

Specifically, we are 95% confident that the mean difference of fugitive methane emissions is between .2301867 and .6960284 million metric tons.

Question 7. Given an entity's total co2 emissions or fugitive methane emissions, predict whether it came from a natural gas or oil & ngl plant.

```
emissions_data3 <- emissions_data_24 %>% filter(commodity == "Natural Gas" | commodity == "Oil & NGL")

mean_length <- mean(emissions_data3$total_emissions_MtCO2e)
sd_length <- sd(emissions_data3$total_emissions_MtCO2e)
emissions_data3$norm_co2 <- (emissions_data3$total_emissions_MtCO2e-mean_length)/sd_length

set.seed(10)

training_indices = createDataPartition(emissions_data3$norm_co2,p = 0.80,list = FALSE)

train_set <- emissions_data3[training_indices, ]
test_set <- emissions_data3[-training_indices, ]

naive_classifier_co2 = train(form = commodity ~ norm_co2,
                            data = train_set,
                            trControl = trainControl(method = "cv", number = 5),
                            method = "naive_bayes")

naive_classifier_co2
```

```
## Naive Bayes
##
## 144 samples
##   1 predictor
##   2 classes: 'Natural Gas', 'Oil & NGL'
##
## No pre-processing
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 115, 115, 116, 115, 115
## Resampling results across tuning parameters:
##
##   usekernel  Accuracy   Kappa
##   FALSE      0.5270936   0.03207129
##    TRUE      0.4788177  -0.06423903
##
## Tuning parameter 'laplace' was held constant at a value of 0
## Tuning
##  parameter 'adjust' was held constant at a value of 1
## Accuracy was used to select the optimal model using the largest value.
## The final values used for the model were laplace = 0, usekernel = FALSE
##  and adjust = 1.
```

A Kappa value of 0.032 shows that there is poor agreement between the true and predicted values, so this is not a reliable classifier.

```
mean_length <- mean(emissions_data3$fugitive_methane_emissions_MtCH4)
sd_length <- sd(emissions_data3$fugitive_methane_emissions_MtCH4)
emissions_data3$norm_methane <- (emissions_data3$fugitive_methane_emissions_MtCH4-mean_length)/sd_lengtl
```

```r
set.seed(10)

training_indices = createDataPartition(emissions_data3$norm_methane,p = 0.80,list = FALSE)

train_set <- emissions_data3[training_indices, ]
test_set <- emissions_data3[-training_indices, ]

naive_classifier_methane = train(form = commodity ~ norm_methane,
                                 data = train_set,
                                 trControl = trainControl(method = "cv", number = 5),
                                 method = "naive_bayes")

naive_classifier_methane
```

```
## Naive Bayes
##
## 144 samples
##   1 predictor
##   2 classes: 'Natural Gas', 'Oil & NGL'
##
## No pre-processing
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 115, 116, 115, 115, 115
## Resampling results across tuning parameters:
##
##   usekernel  Accuracy   Kappa
##   FALSE      0.6110837  0.2071119
##    TRUE      0.5770936  0.1452654
##
## Tuning parameter 'laplace' was held constant at a value of 0
## Tuning
##  parameter 'adjust' was held constant at a value of 1
## Accuracy was used to select the optimal model using the largest value.
## The final values used for the model were laplace = 0, usekernel = FALSE
##  and adjust = 1.
```

A Kappa value of 0.207 shows that there is fair agreement between the true and predicted values, so this classifier has some slight predictive power. Methane is more reliable than co2 emissions for identifying the plant types.

```r
#classifier with multiple variables

emissions_data3 <- emissions_data_24 %>% filter(commodity == "Natural Gas" | commodity == "Oil & NGL")

mean_co2 <- mean(emissions_data3$total_emissions_MtCO2e)
sd_co2 <- sd(emissions_data3$total_emissions_MtCO2e)
emissions_data3$norm_co2 <- (emissions_data3$total_emissions_MtCO2e-mean_co2)/sd_co2

mean_methane <- mean(emissions_data3$fugitive_methane_emissions_MtCH4)
sd_methane <- sd(emissions_data3$fugitive_methane_emissions_MtCH4)
emissions_data3$norm_methane <- (emissions_data3$fugitive_methane_emissions_MtCH4-mean_methane)/sd_metha
```

```r
set.seed(10)

training_indices = createDataPartition(emissions_data3$norm_co2,p = 0.80,list = FALSE)

train_set <- emissions_data3[training_indices, ]
test_set <- emissions_data3[-training_indices, ]

naive_classifier_co2 = train(form = commodity ~ norm_co2+norm_methane+parent_type,
                             data = train_set,
                             trControl = trainControl(method = "cv", number = 5),
                             method = "naive_bayes")
```

```r
test_set$commodity <- as.factor(test_set$commodity)

naive_classifier_final <- naive_classifier_co2$finalModel
new_emissions <- test_set[, c("norm_co2", "norm_methane", "parent_type")]
predicted_plant <- predict(naive_classifier_final, new_emissions)
```

```
## Warning: predict.naive_bayes(): only 2 feature(s) out of 3 defined in the naive_bayes object "naive_
```

```r
cm_details <- confusionMatrix(data = predicted_plant,reference = test_set$commodity)

cm_details
```

```
## Confusion Matrix and Statistics
##
##              Reference
## Prediction    Natural Gas Oil & NGL
##   Natural Gas           7         0
##   Oil & NGL            10        18
##
##                Accuracy : 0.7143
##                  95% CI : (0.537, 0.8536)
##     No Information Rate : 0.5143
##     P-Value [Acc > NIR] : 0.013006
##
##                   Kappa : 0.4186
##
##  Mcnemar's Test P-Value : 0.004427
##
##             Sensitivity : 0.4118
##             Specificity : 1.0000
##          Pos Pred Value : 1.0000
##          Neg Pred Value : 0.6429
##              Prevalence : 0.4857
##          Detection Rate : 0.2000
##    Detection Prevalence : 0.2000
##       Balanced Accuracy : 0.7059
##
##        'Positive' Class : Natural Gas
##
```

71.43% accuracy with a kappa value of 0.4186 shows there is moderate agreement and is pretty accurate at making predictions.